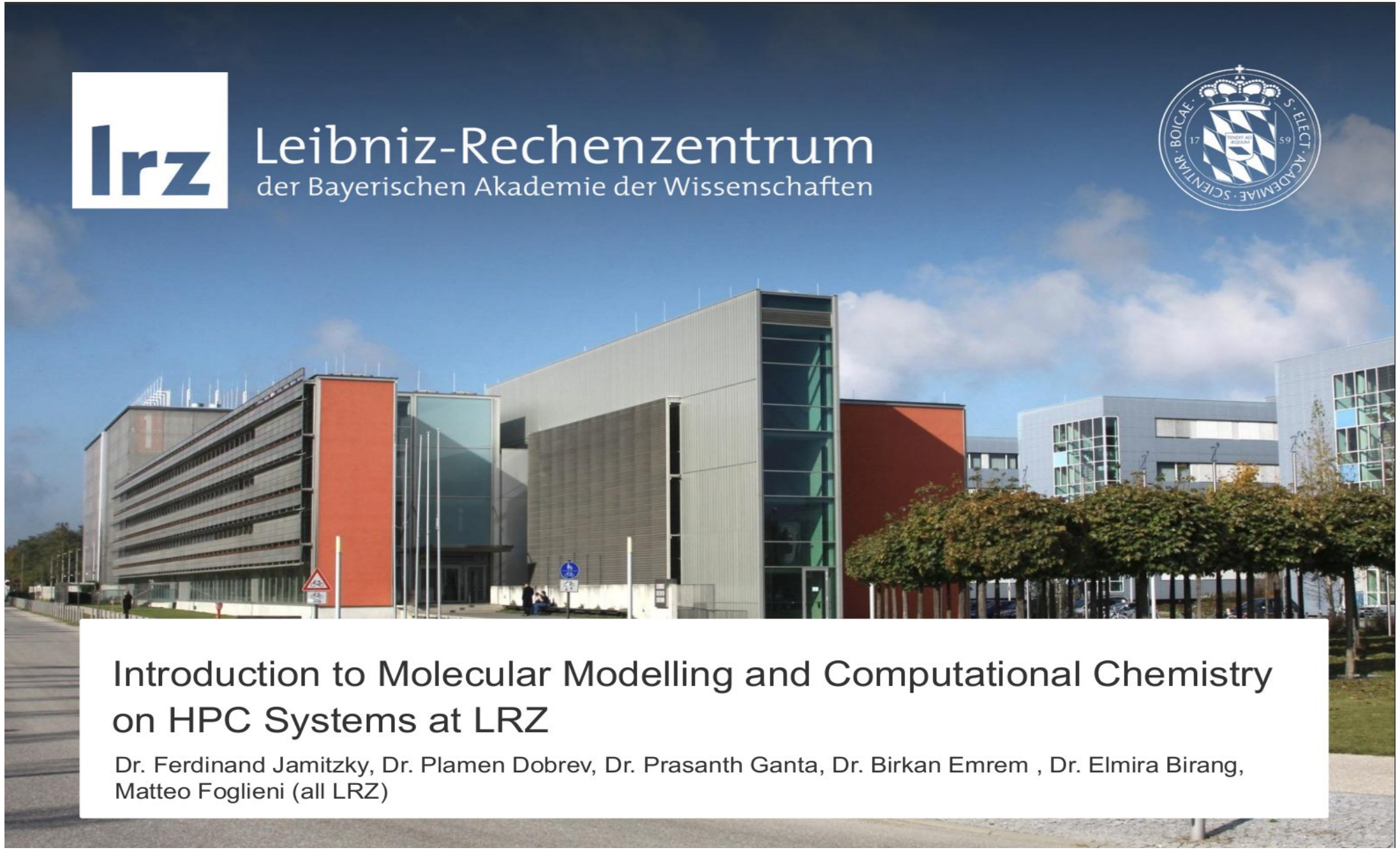
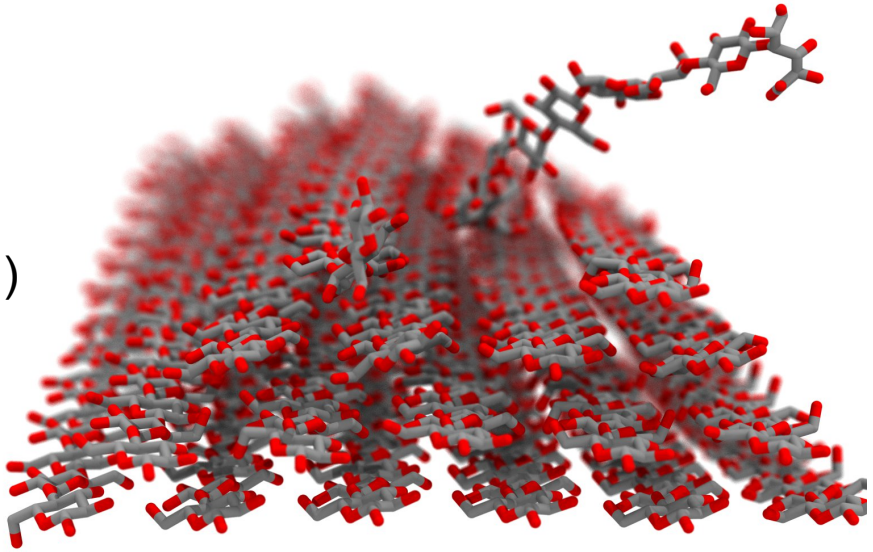


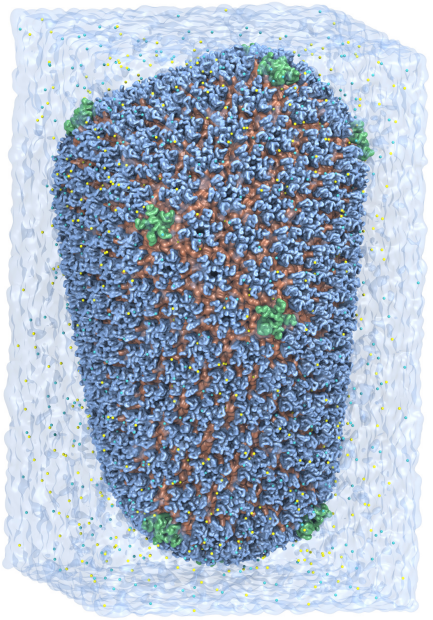

# Introduction to Molecular Modelling and Computational Chemistry on HPC Systems at LRZ

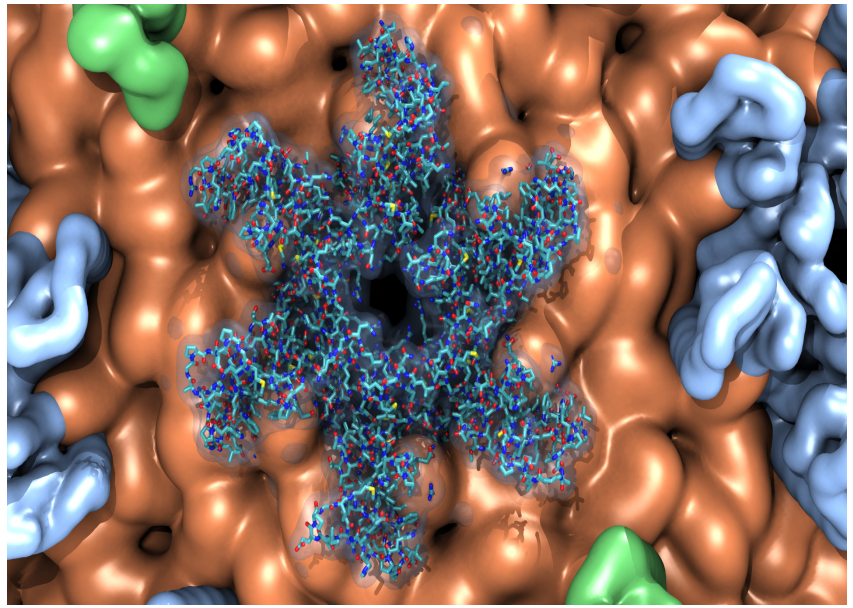
Dr. Ferdinand Jamitzky, Dr. Plamen Dobrev, Dr. Prasanth Ganta, Dr. Birkan Emrem , Dr. Elmira Birang, Matteo Foglieni (all LRZ)

# Introduction to VMD (Visual Molecular Dynamics)

- Visualizing large Molecular Dynamics Trajectories
- Supports many molecular formats (PDB, PSF, DCD, etc.)
- Commonly used with NAMD and CHARMM
- Open-source and cross-platform
- Data Analysis tools
- HD images and Movie making
- Built in tcl and python scripting







Humphrey, W., Dalke, A. and Schulten, K., `VMD - Visual Molecular Dynamics', J. Molec. Graphics 1996, 14.1, 33-38.

# Installing VMD on Linux OS

## 1. Go to Download: www.ks.uiuc.edu/Research/vmd

**Version 2.0.0 LATEST ALPHA (2025-03-13) Platforms:**

Latest pre-release ALPHA test version

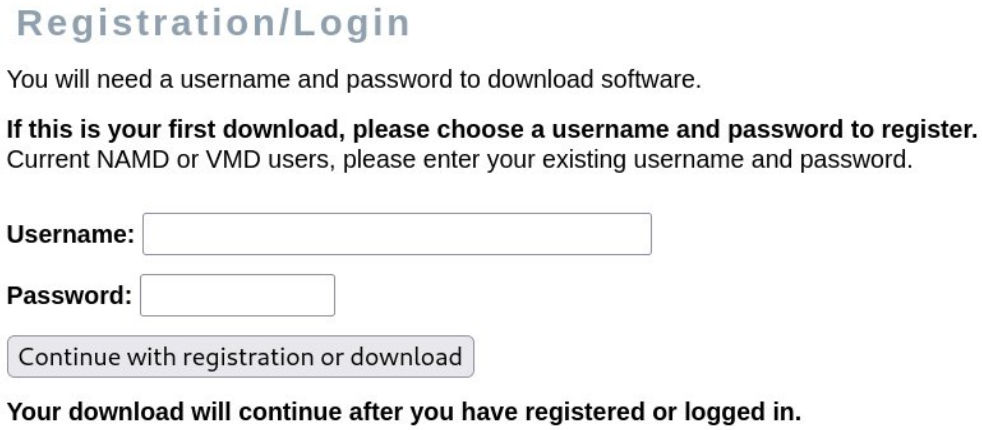
- **LINUX_64 (RHEL 8+) OpenGL, CUDA, OptiX RTX, RTX RTF**

**Version 1.9.4 LATEST ALPHA (2023-06-08) Platforms:**

Latest pre-release ALPHA test version

- **Source Code**
- **LINUX_64 (RHEL 7+) OpenGL, CUDA, OptiX RTX, OSPRay**
- **LINUX_64 (RHEL 7+) OpenGL, CUDA, OptiX RTX, OSPRay,**
- **MacOS 11.x, ARM64 (64-bit "M1" Macs)** (Apple MacOS-X 11
- **MacOS 10.15, x86_64 (64-bit Intel x86_64)** (Apple MacOS-X
- **Windows 64-bit, CUDA, OptiX, OSPRay (64-bit Intel x86_64**

## 2. Register




## 3. VMD folder contents

```
di97gef@badwlrz-cl18014:vmd-1.9.4a57$ ls
Announcement        distrib     msvc      scripts
bin                 doc         plugins   shaders
configure           lib         proteins  src
configure.options   LICENSE     python
data                LINUXAMD64  README
```

## 4. modify configure file (optional)

```
##############################################################################
# User modifiable installation parameters, can be overridden by env variables
##############################################################################
# Name of shell script used to start program; this is the name used by users
$install_name = "vmd";

# Directory where VMD startup script is installed, should be in users' paths.
$install_bin_dir="/home/di97gef/.local/bin/";

# Directory where VMD files and executables are installed
$install_library_dir="/home/di97gef/Documents/packages/vmd/";
```

# Installing VMD on Linux OS

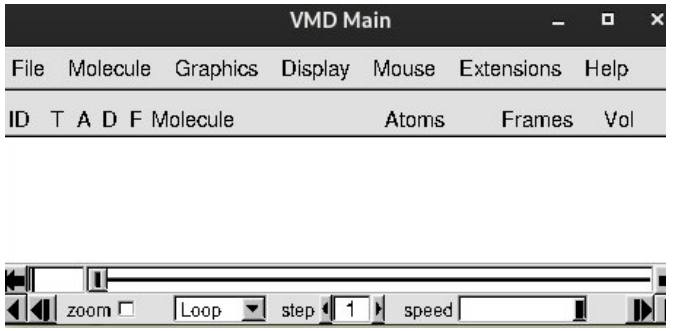
```
# Architectures:
# ------------------------------------------------------------
# AIX6_64       - compile on an RS/6000 machine (gcc/g++/xlc/xlC)
# ANDROIDARMV7A - cross-compile for Android on ARM V7A
# BLUEWATERS    - compile for NSF/NCSA Blue Waters
# CRAY_XC       - compile for Cray XCx0 machines with GPUs
# CRAY_XK       - compile for Cray XKx machines with GPUs
# FREEBSD       - compile on a FreeBSD machine (gcc/g++)
# FREEBSDAMD64  - compile on a FreeBSD machine (gcc/g++)
# LINUX         - compile on a Linux machine (x86 32-bit)
# LINUXAMD64    - compile on a Linux machine (x86 64-bit AMD64 or EM64T)
# LINUXARM64    - compile on a Linux machine (64-bit ARM)
# LINUXPPC64    - compile on a Linux machine (64-bit PowerPC)
# MACOSX        - compile on a MacOS-X PowerPC machine
# MACOSXARM64   - compile on a MacOS-X ARM64 machine, 64-bit ABI
# MACOSXX86     - compile on a MacOS-X x86 machine, 32-bit ABI
# MACOSXX86_64  - compile on a MacOS-X x86 machine, 64-bit ABI
```

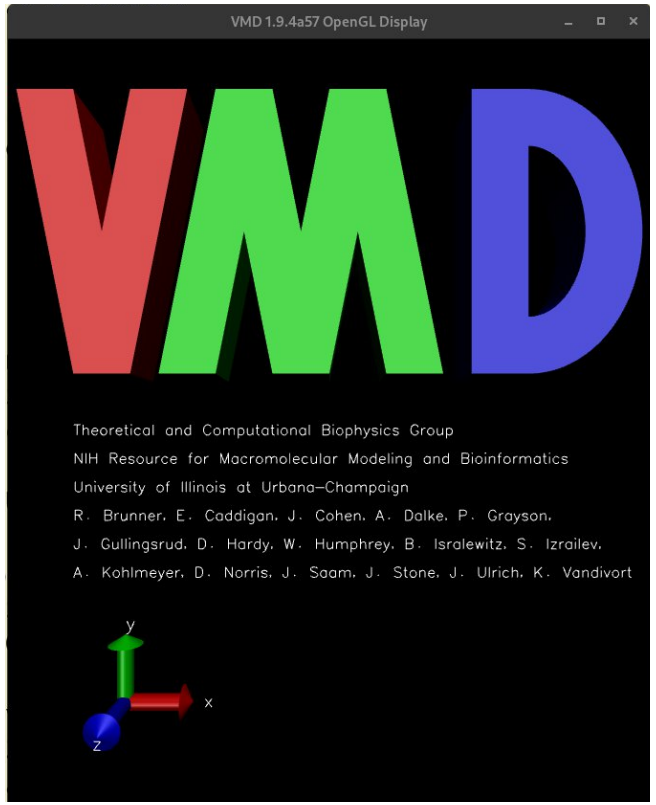
5. Run configure

```
di97gef@badwlrz-cl18014:vmd-1.9.4a57$ ./configure LINUXAMD64
```

6. Go to src folder and vmd install

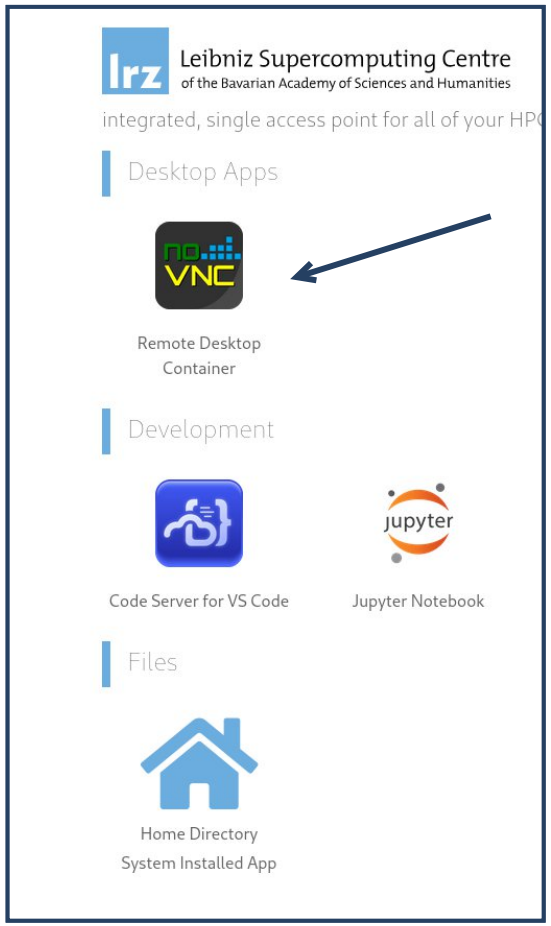
```
di97gef@badwlrz-cl18014:src$ make install
if [ ! -d "/home/di97gef/Documents/packages/vmd/" ]; then \
        mkdir -p "/home/di97gef/Documents/packages/vmd/" ; \
fi ; \
if [ ! -d "/home/di97gef/.local/bin/" ]; then \
        mkdir -p "/home/di97gef/.local/bin/" ; \
fi ; \
if [ ! -d "/home/di97gef/Documents/packages/vmd/"/doc ]; then \
        mkdir -p "/home/di97gef/Documents/packages/vmd/"/doc; \
fi
```

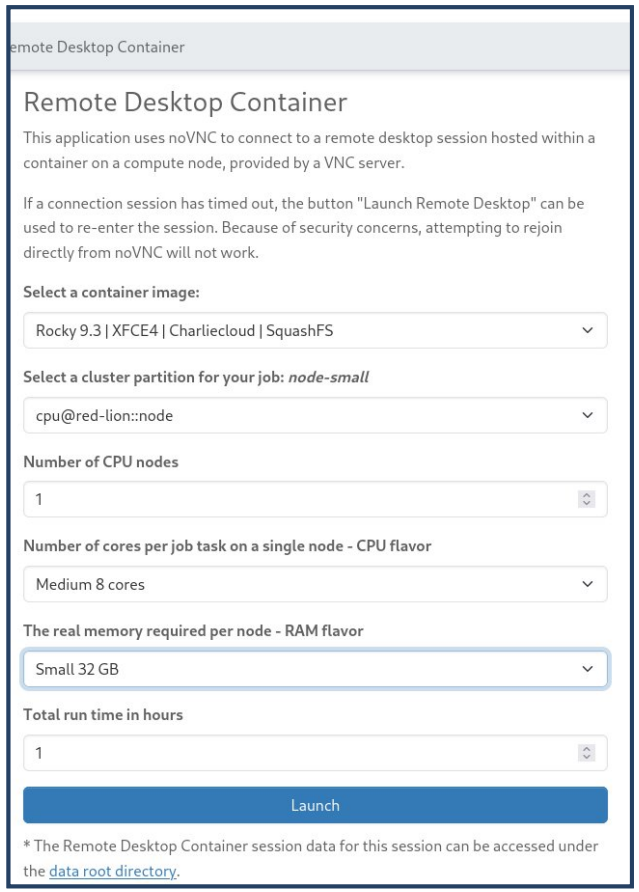


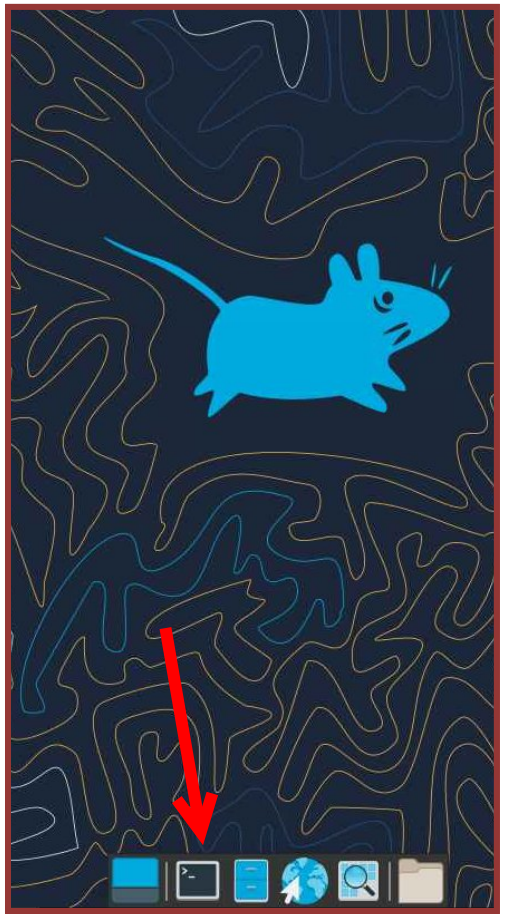



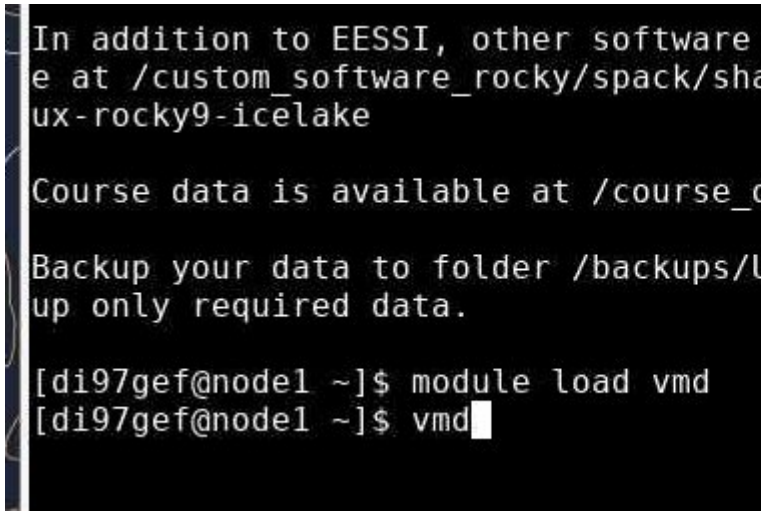
# VMD in this course

https://ood.red-lion.gcs-devcloud.hpc.lrz.de:8443/

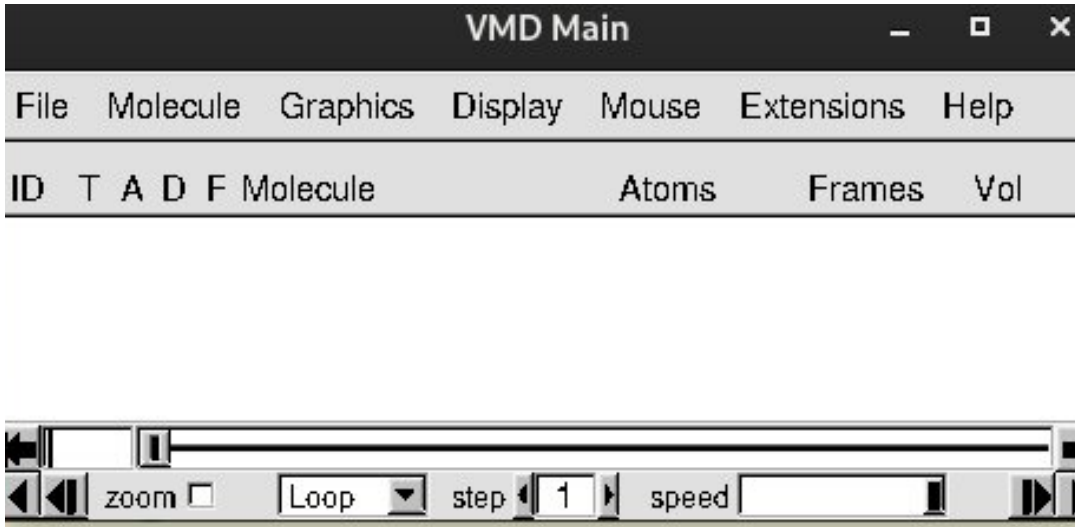


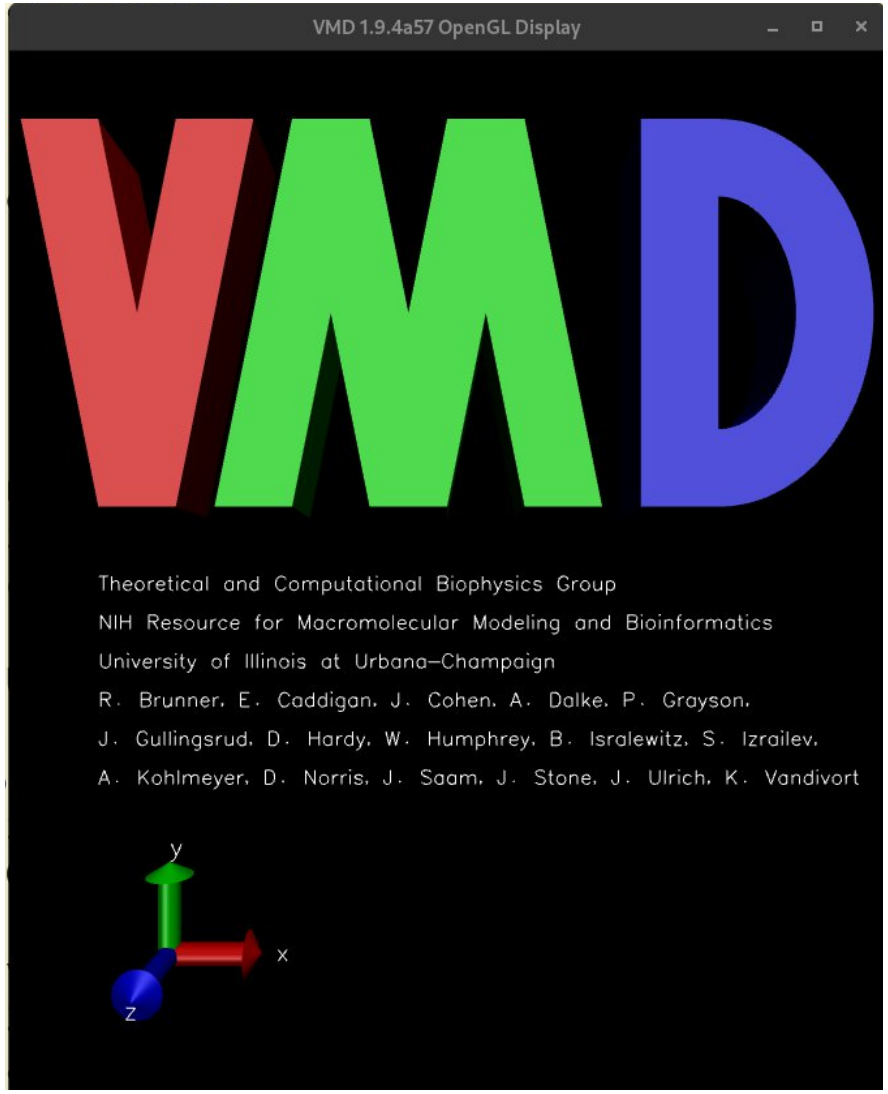



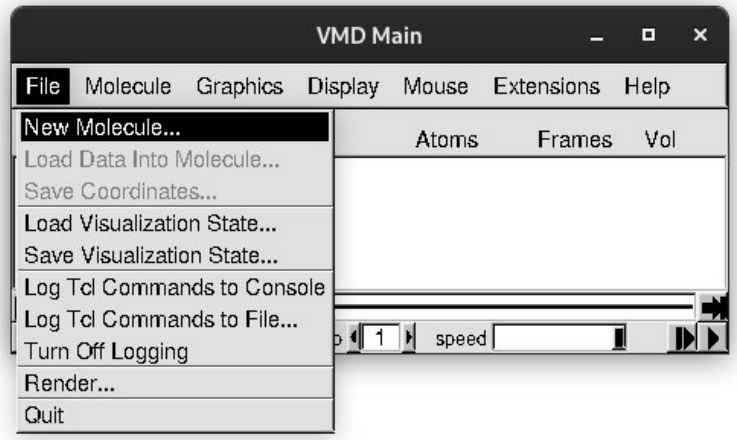


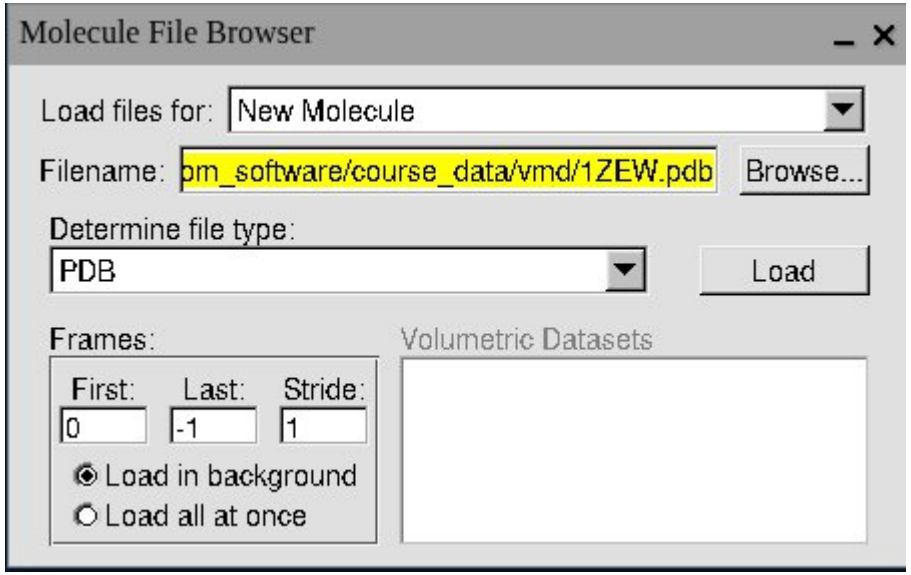
# VMD in this course

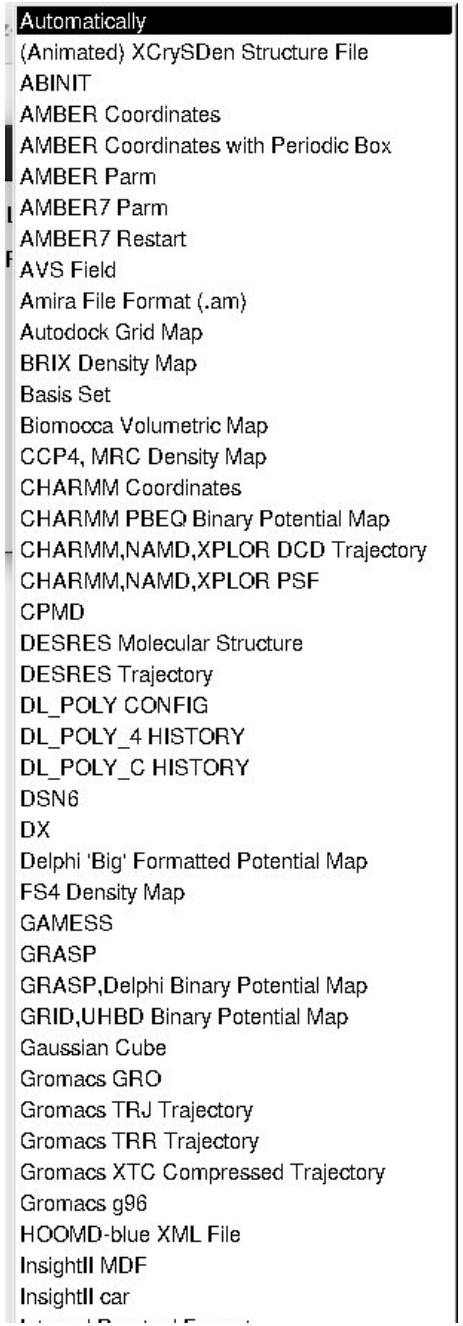


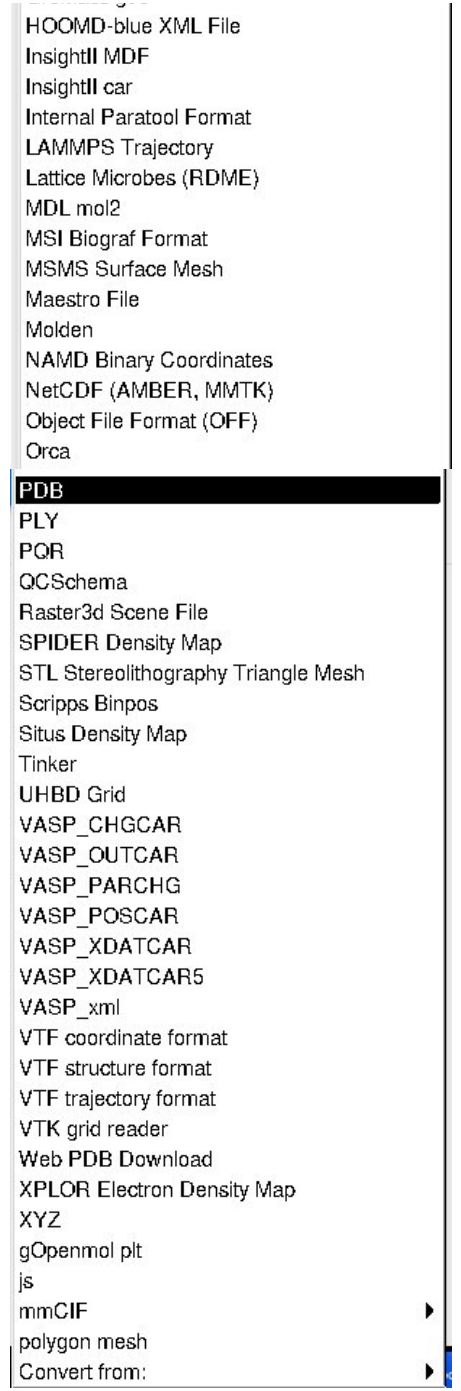



# Supported Formats



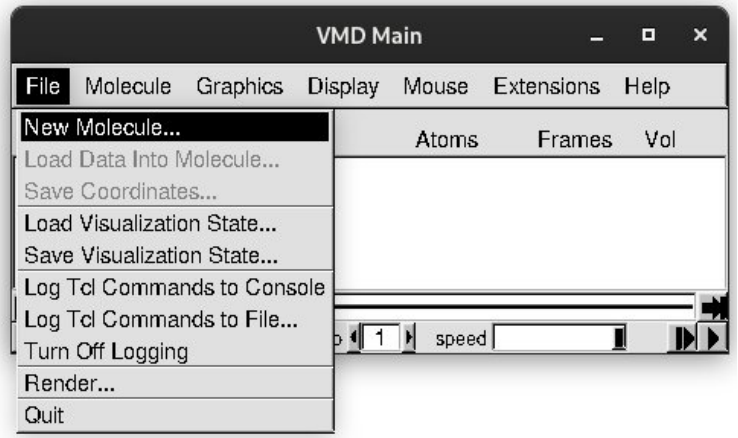



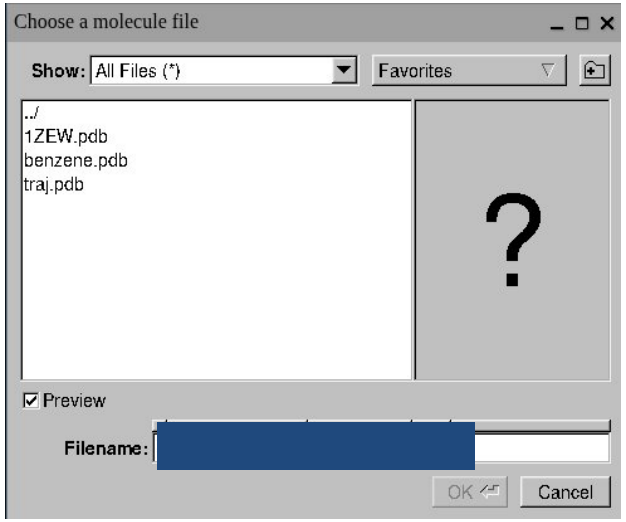



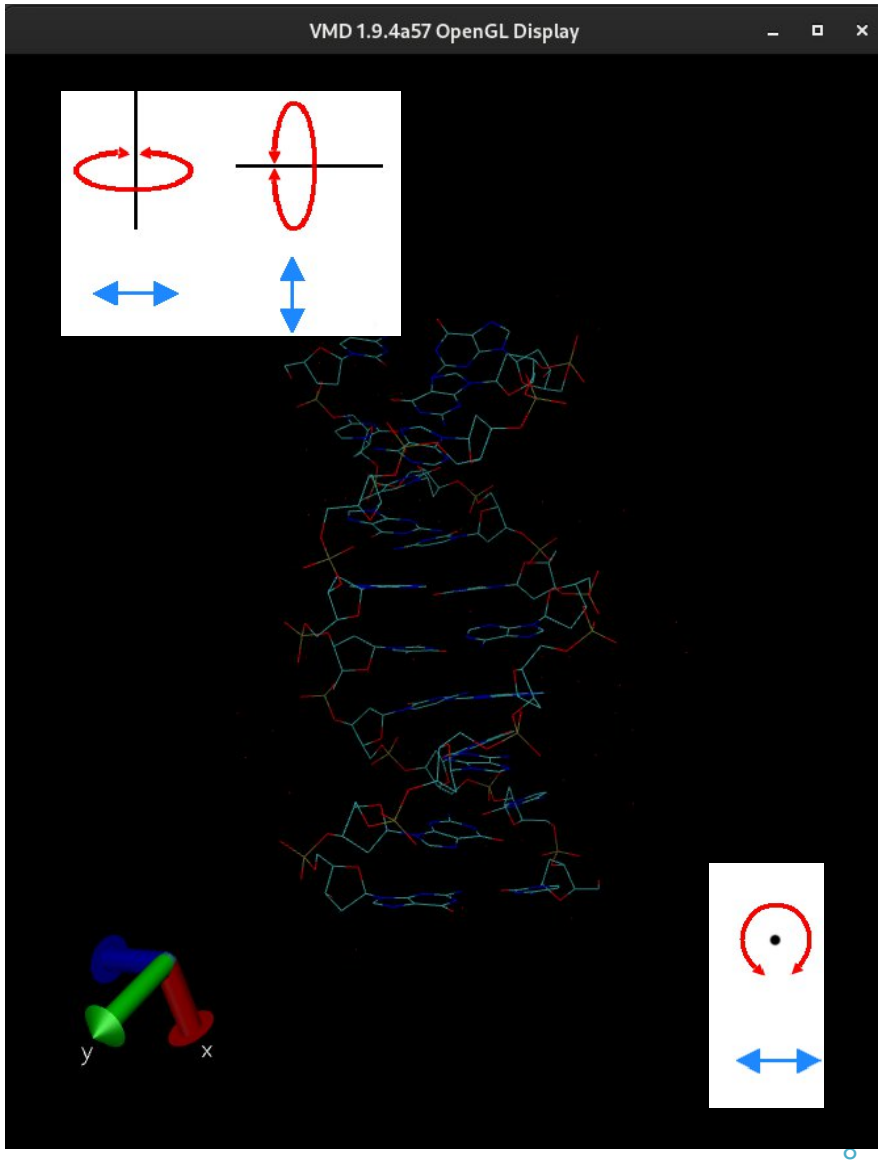



# Interacting with molecule

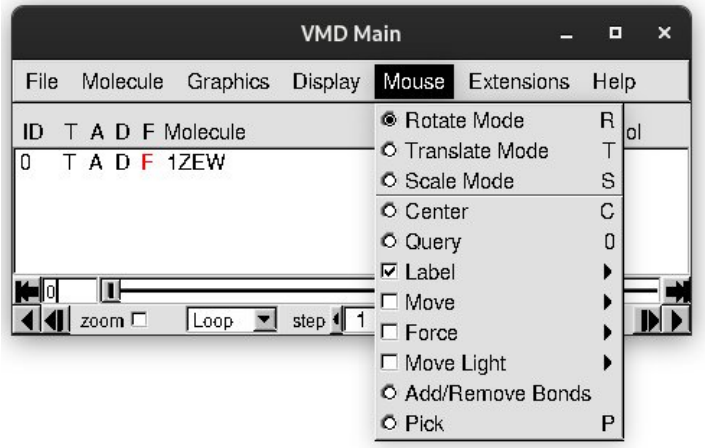
/course_data/BioPhyCourse/2026/02_Visualisation_Tools/04_VMD/data/

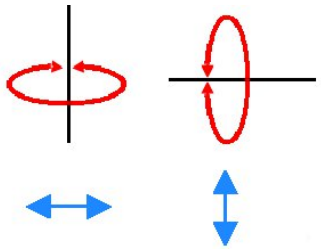



## Rotate, translate and scale



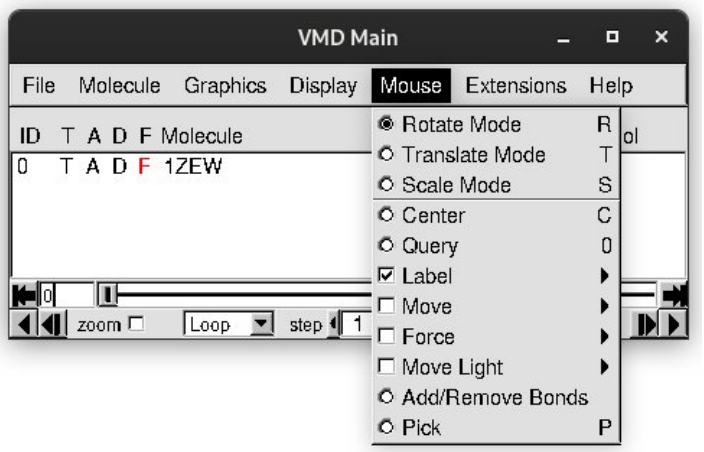
Left Mouse



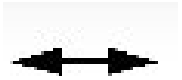
R M

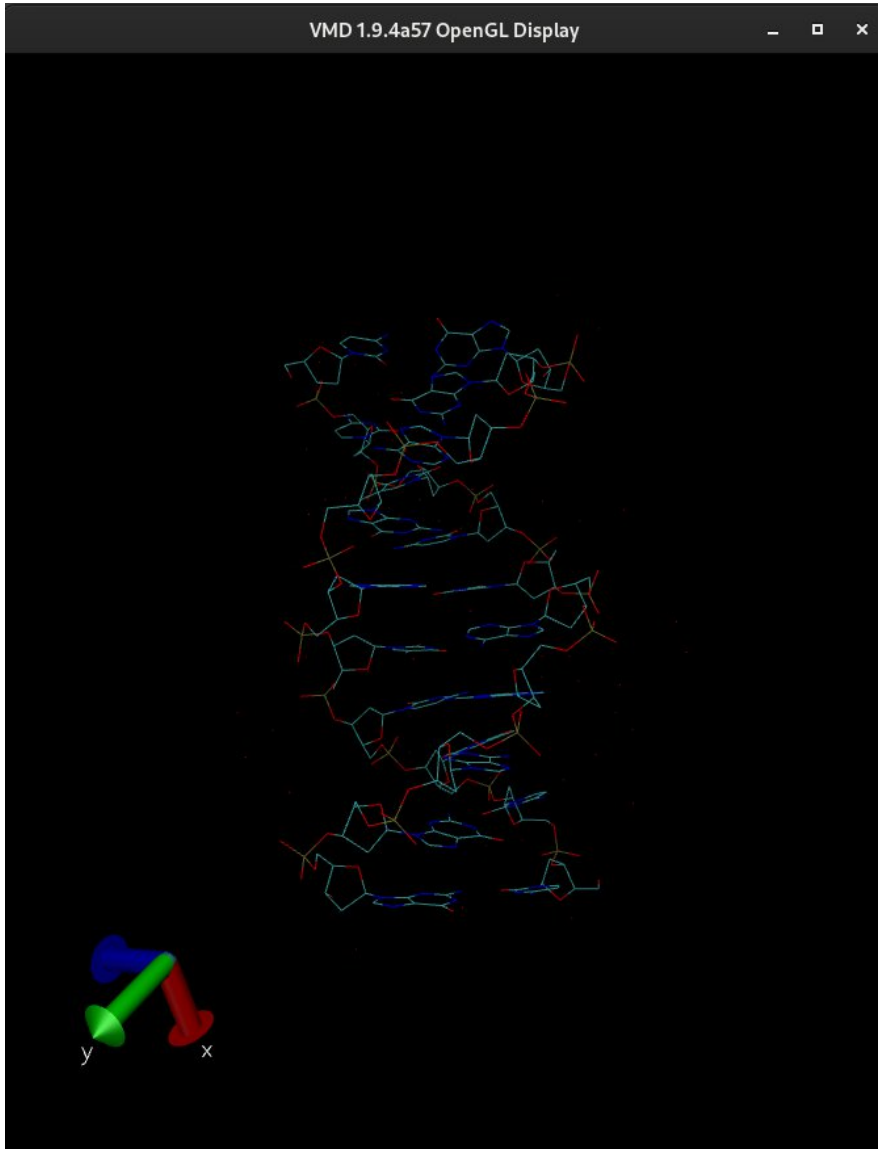
# Interacting with molecule

Translate and scale

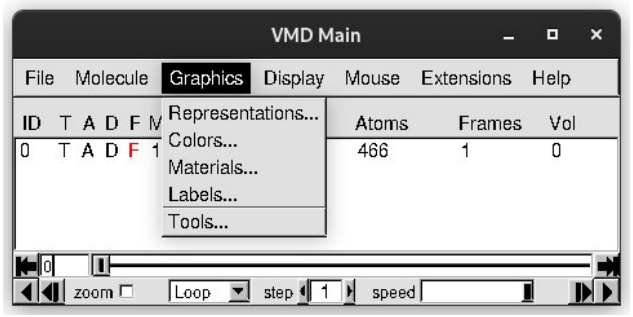


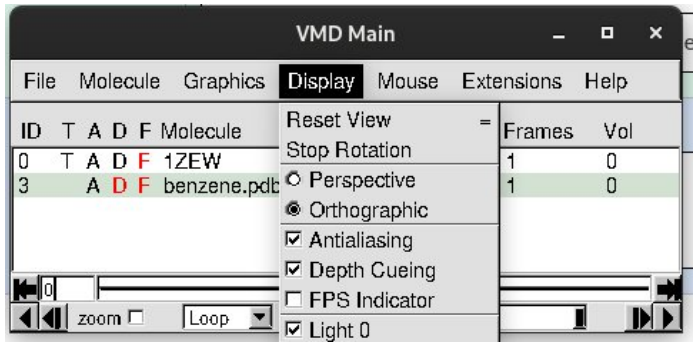

Press T to Translate - 

Press S to Scale - 

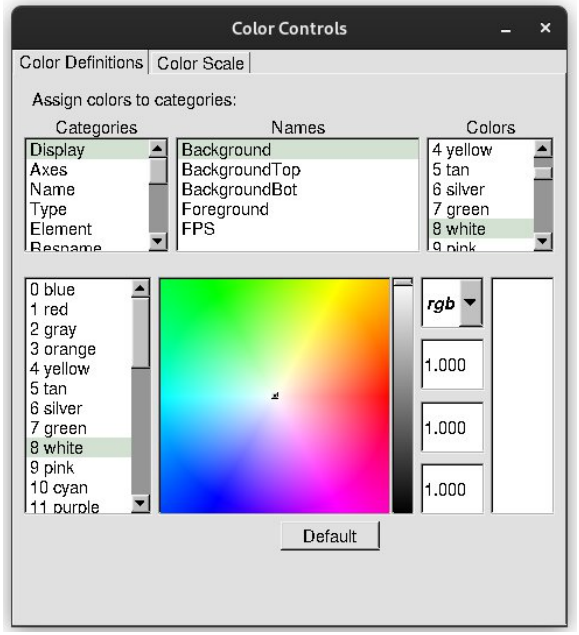


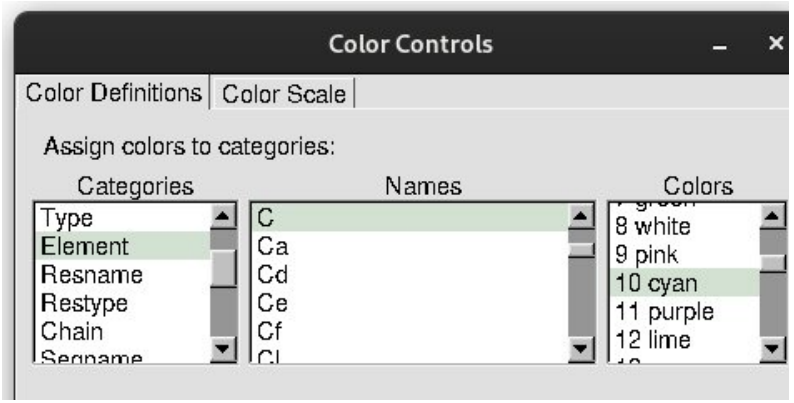
# Color Settings

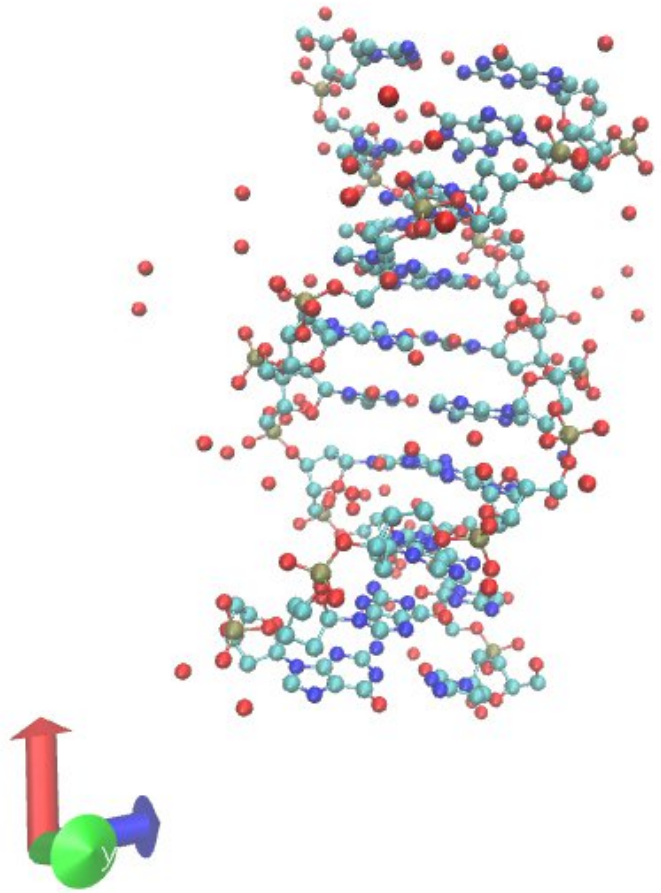


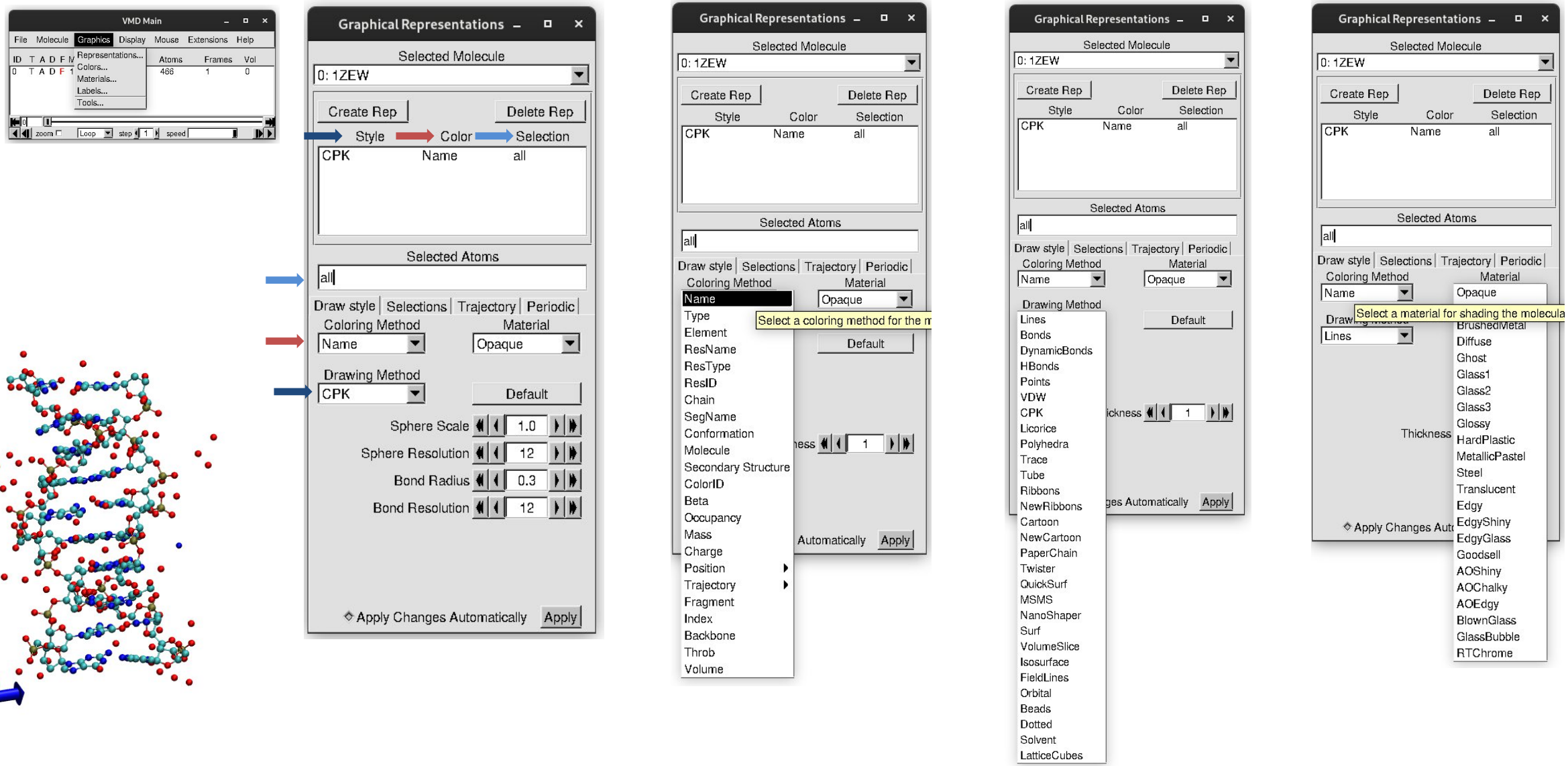

change colors of background, axes, elements.....



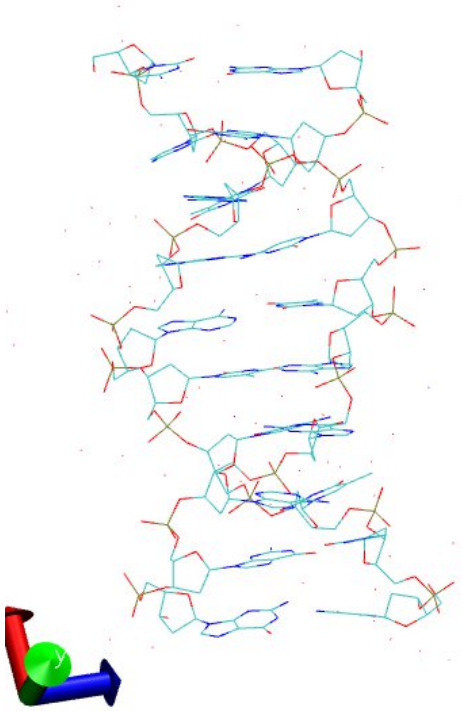



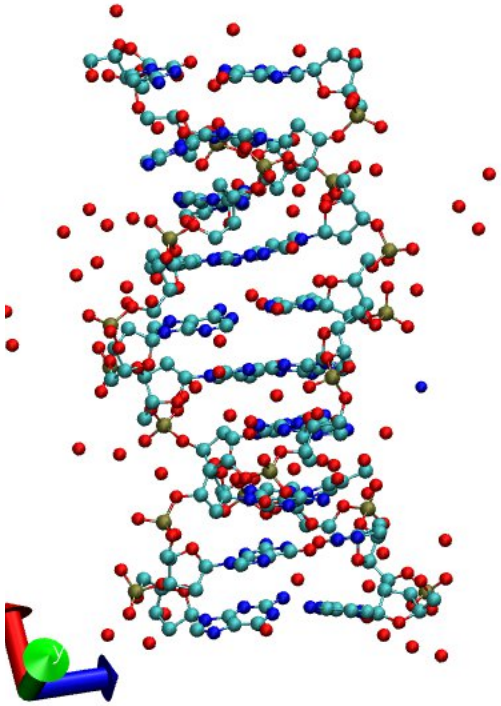
# Graphical Representation



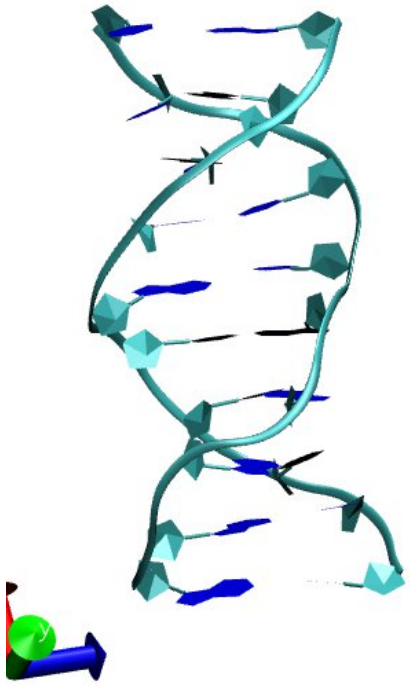
# Graphical Representation: Drawing Method



Lines



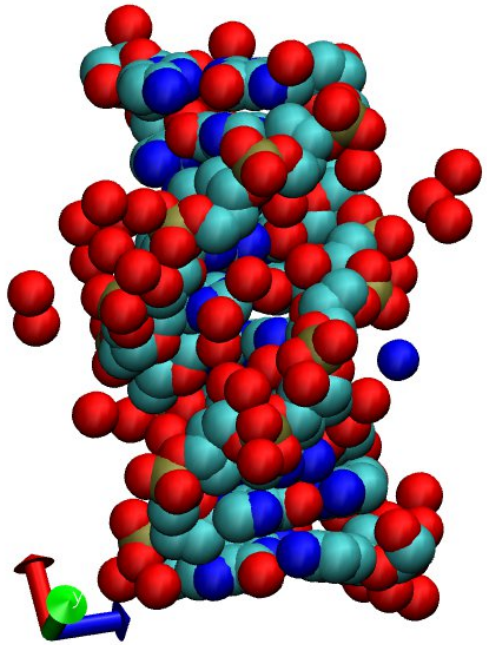
CPK



New Ribbons

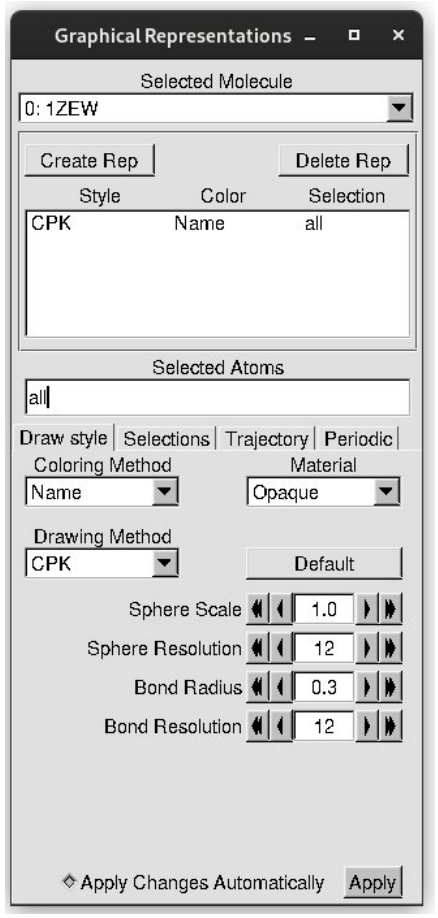


New Cartoon



VDW

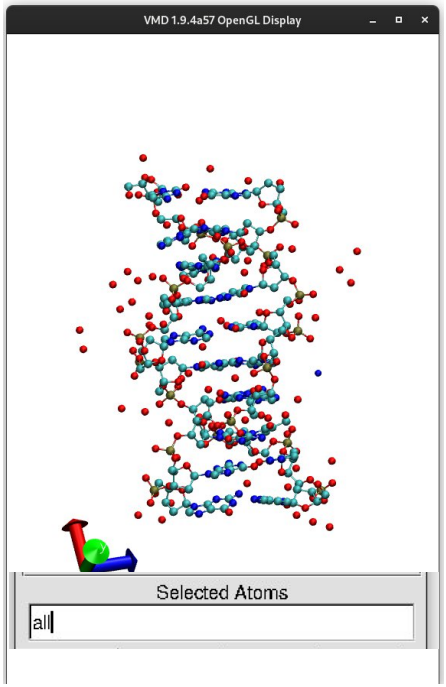
# Graphical Representation: Selected Atoms

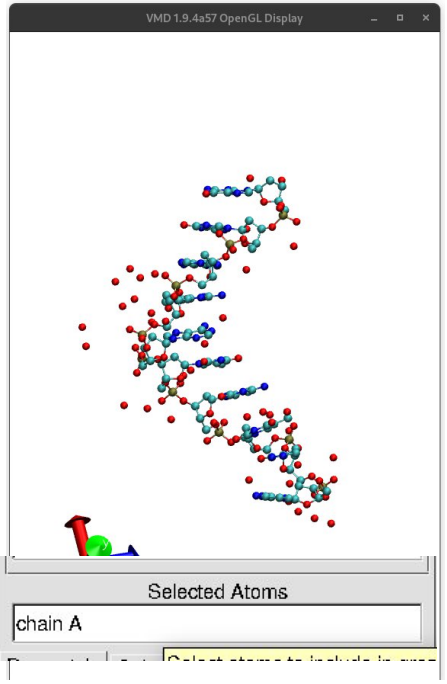


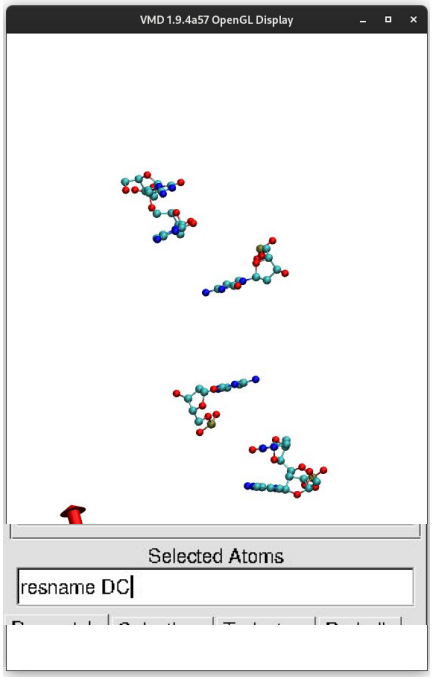

Serial type res chain coordinates

```
ATOM      1  O5'  DC A   1      -0.633   0.238  32.103  1.00 23.11           O
ATOM      2  C5'  DC A   1      -1.560  -0.576  32.829  1.00 22.85           C
ATOM      3  C4'  DC A   1      -1.619  -1.987  32.293  1.00 22.73           C
ATOM      4  O4'  DC A   1      -2.109  -1.980  30.932  1.00
ATOM      5  C3'  DC A   1      -0.282  -2.719  32.253  1.00
```

PDB File



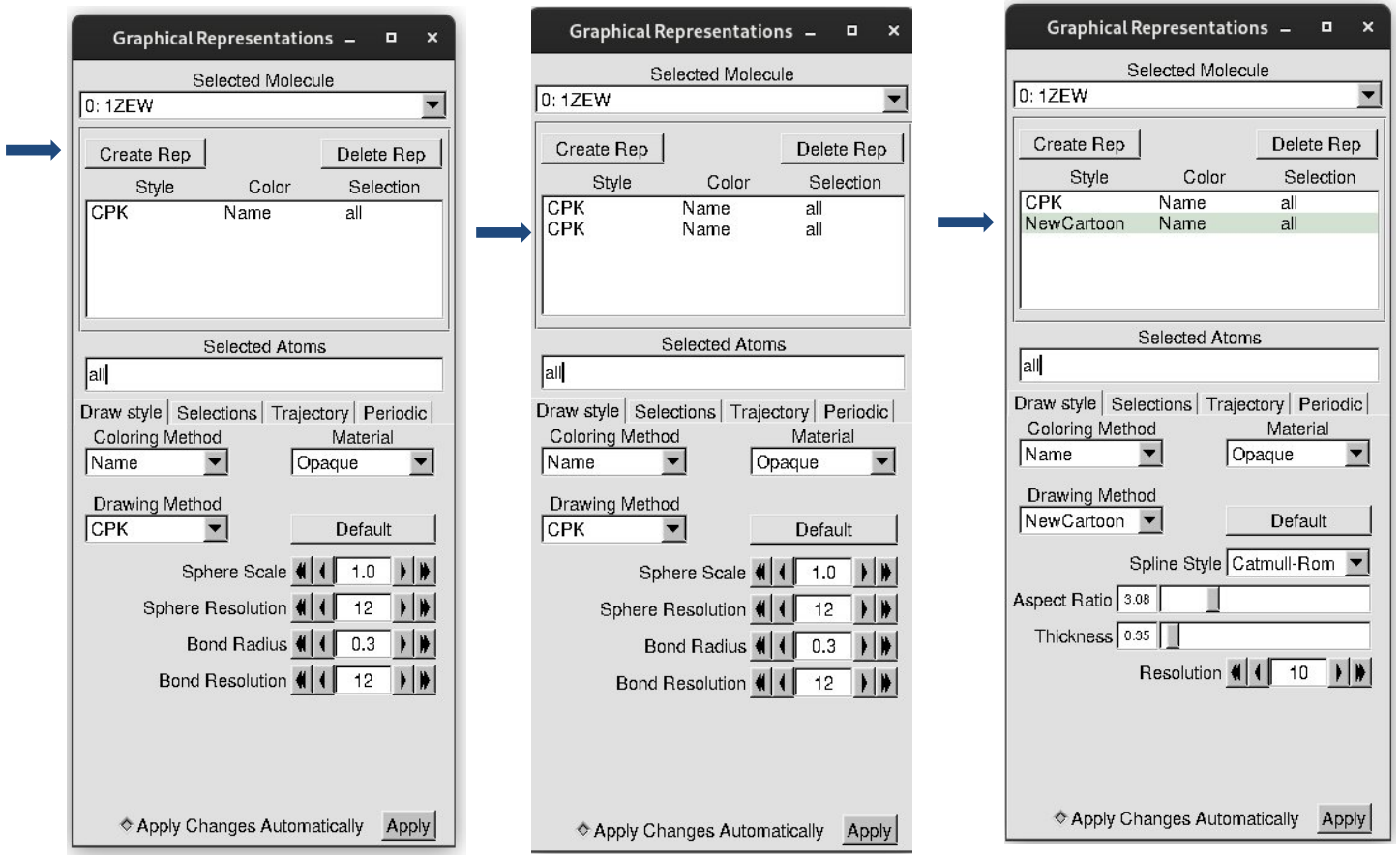

- index 1- 100
- resname DC
- chain A

- x > 10
- y < 20
- x > 10 && y < 20

- element C
- type O5'
- water

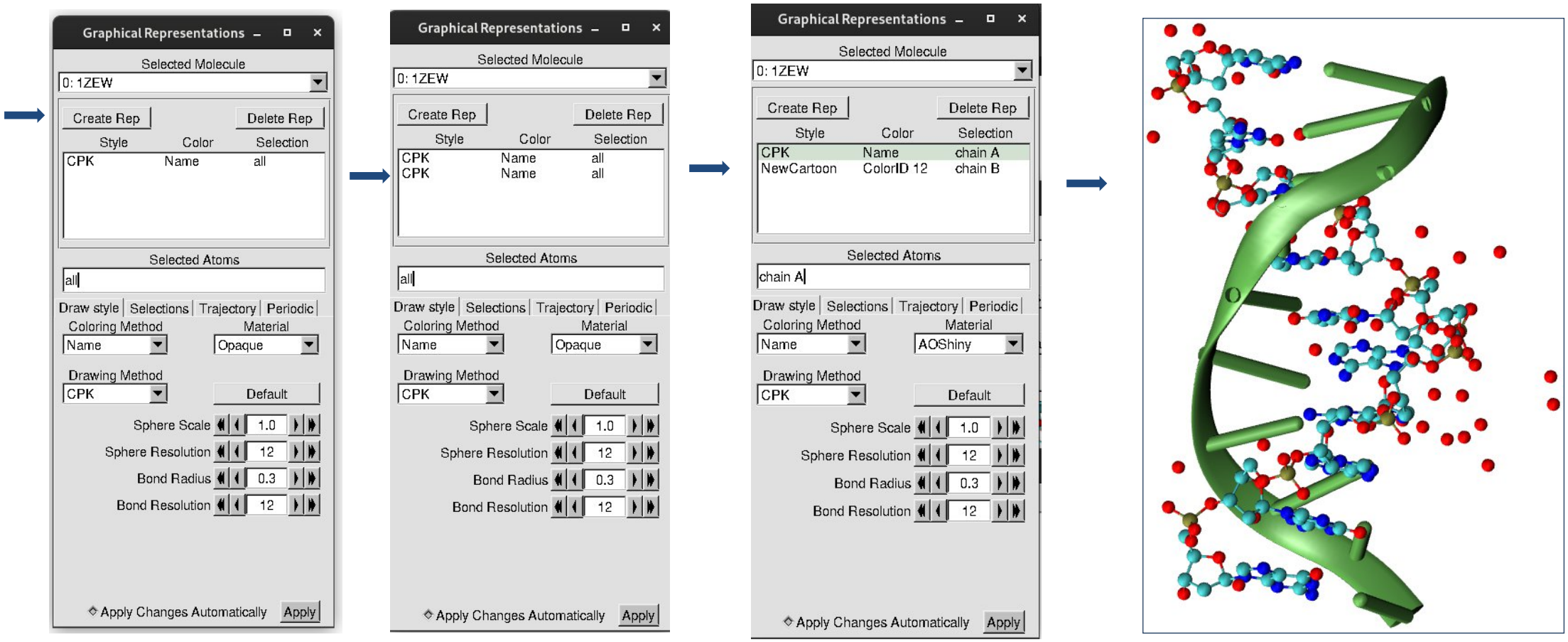
| Columns | Data |
|---|---|
| 1-4 | "ATOM" |
| 7-11[#] | Atom serial number |
| 13-16 | Atom name |
| 17 | Alternate location indicator |
| 18-20[§] | Residue name |
| 22 | Chain identifier |
| 23-26 | Residue sequence number |
| 27 | Code for insertions of residues |
| 31-38 | X orthogonal Å coordinate |
| 39-46 | Y orthogonal Å coordinate |
| 47-54 | Z orthogonal Å coordinate |
| 55-60 | Occupancy |
| 61-66 | Temperature factor |
| 73-76 | Segment identifier[¶] |
| 77-78 | Element symbol |
| 79-80 | Charge |

# Graphical Representation: Create Rep

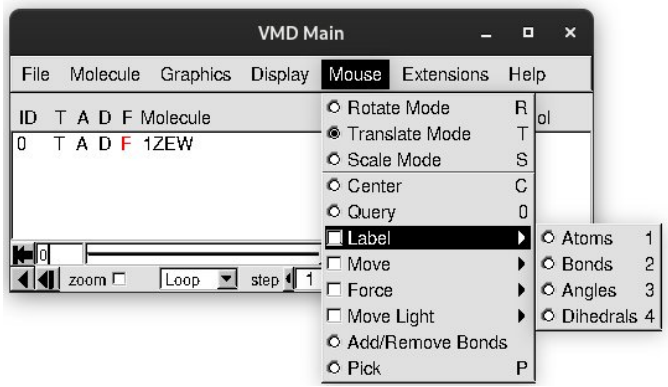




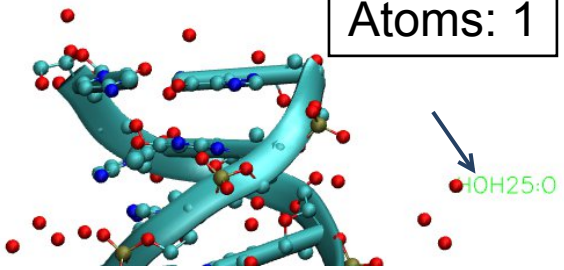

# Graphical Representation: Create Rep



# Label: Atoms, Bonds, Angles and Dihedrals

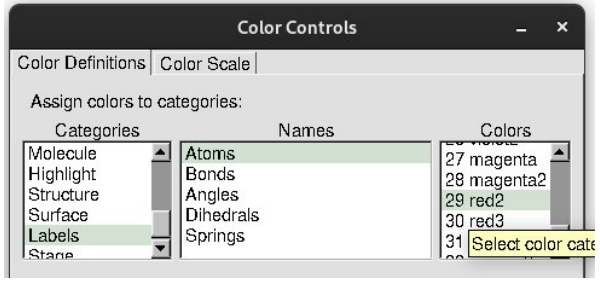



Atoms: 1

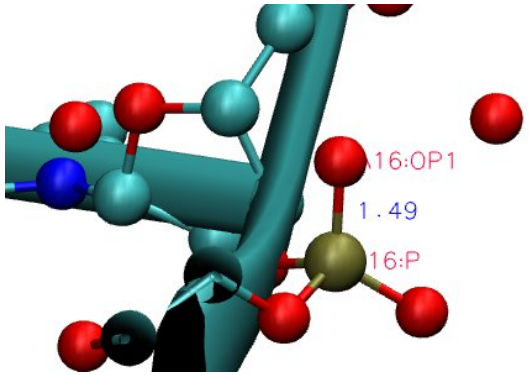



```
Info) picked atom:
Info) ------------
Info) molecule id: 0
Info) trajectory frame: 0
Info) name: O
Info) type: O
Info) index: 436
Info) residue: 52
Info) resname: HOH
Info) resid: 72
Info) chain: A
Info) segname:
Info) x: 18.999001
Info) y: 15.632000
Info) z: 13.699000
Info) Added new Atoms label HOH72:O
```

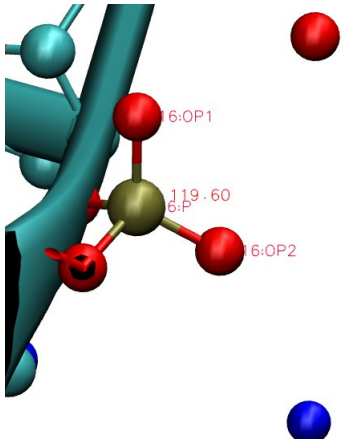
change colors




Bonds: 2

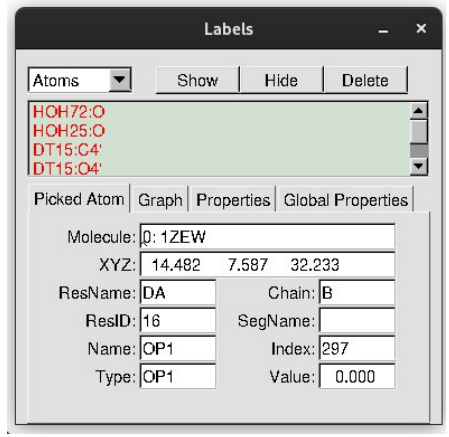



Angles: 3



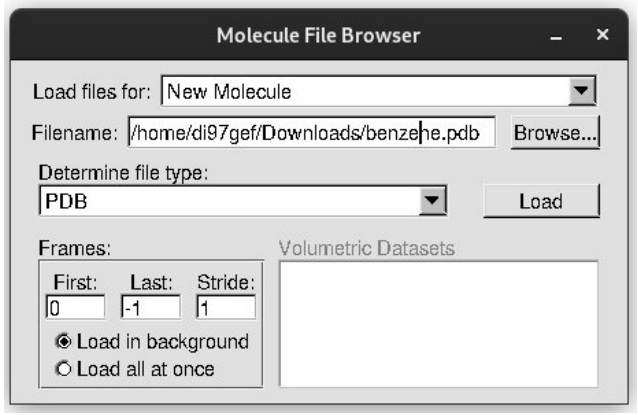

Delete Labels

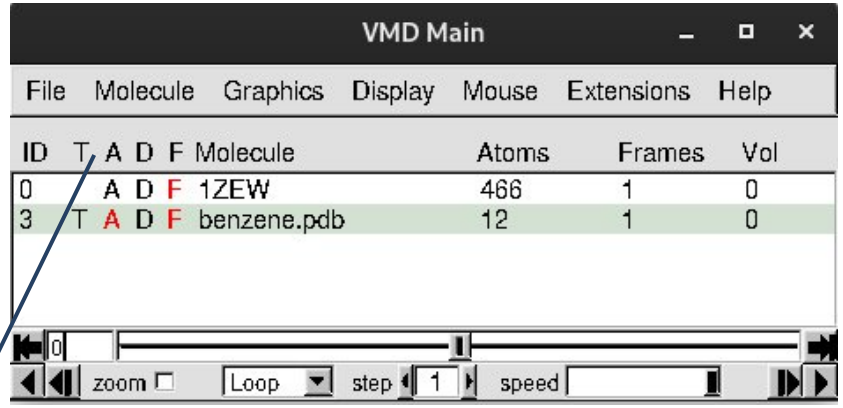
Graphics > Labels



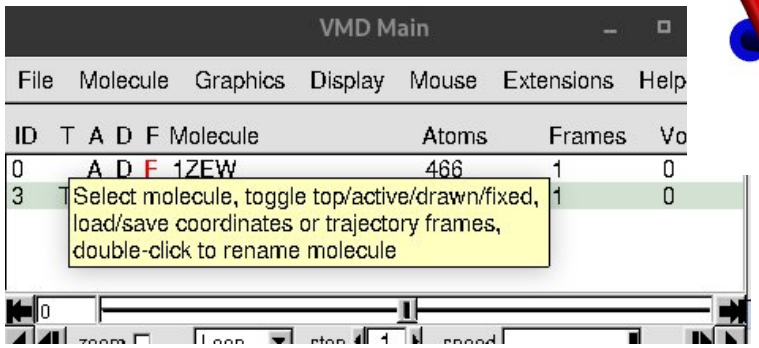
# VMD Main: Multiple Molecules

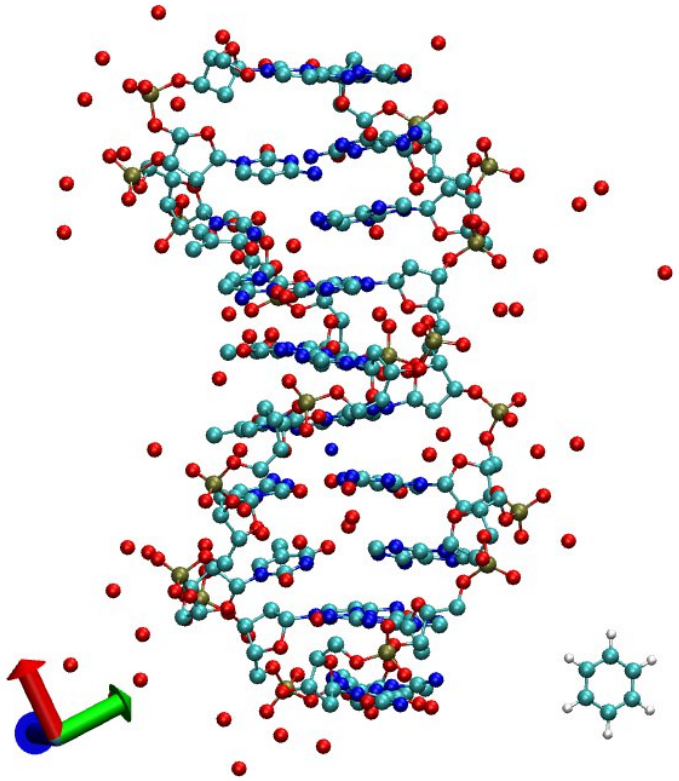
Load new molecule



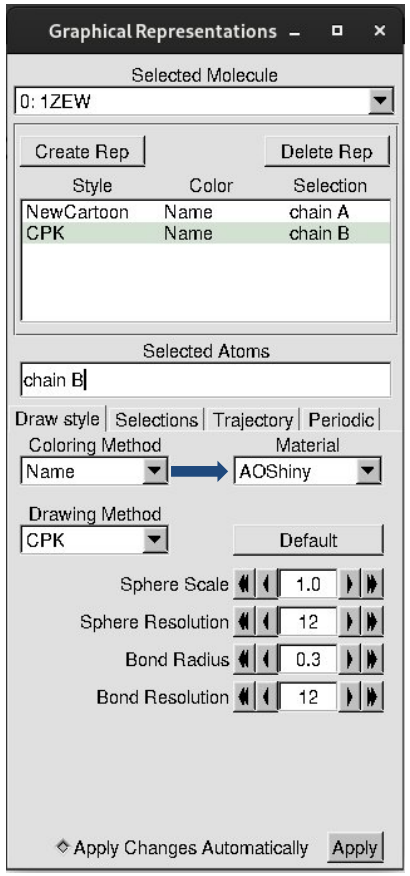

ID - Internal Molecule ID
T - Top; All actions apply to top molecule
A - Active; Trajectory update status
D - Drawn
F - Fixed

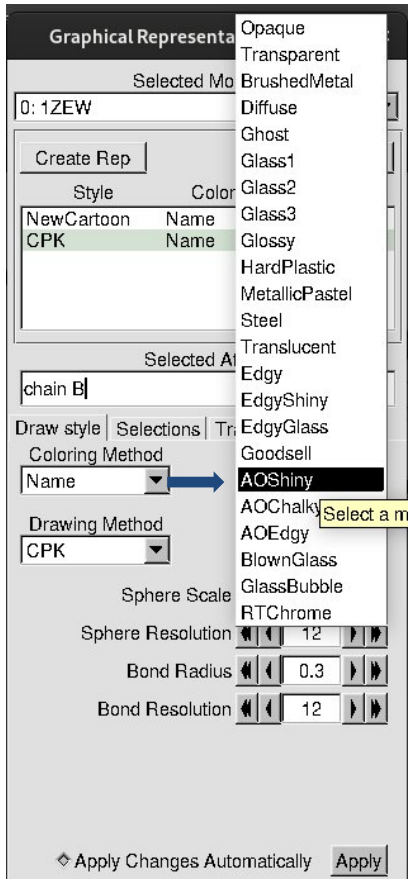


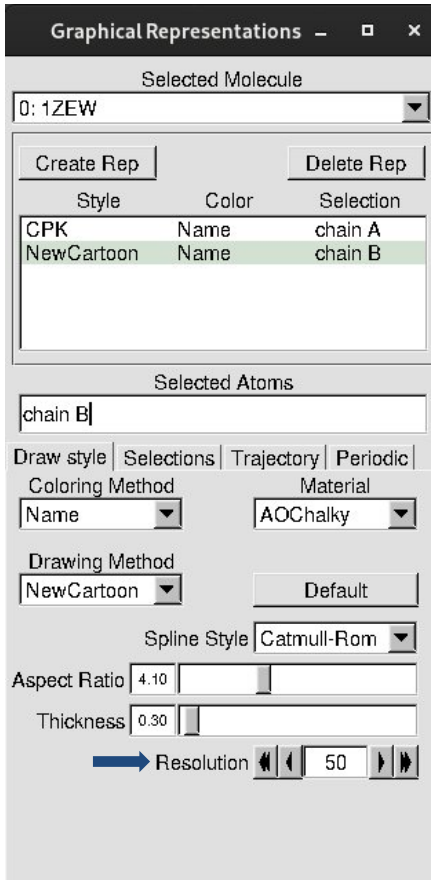
# Rendering: Publication Quality Images

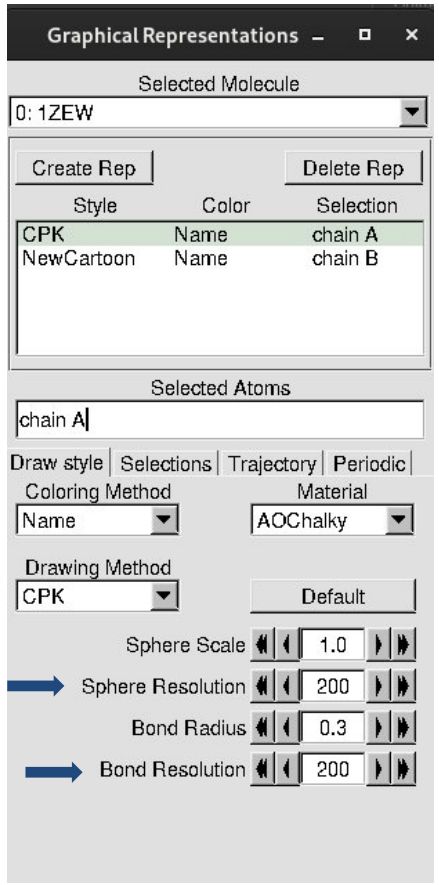
## Change Material Type



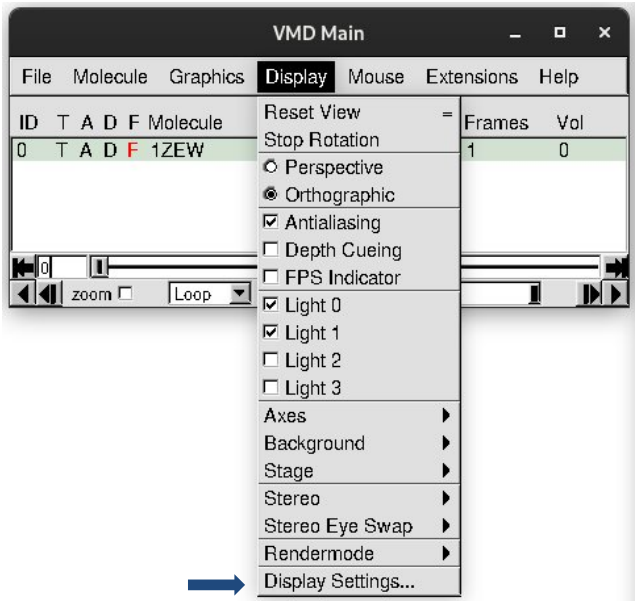

## Change Resolution

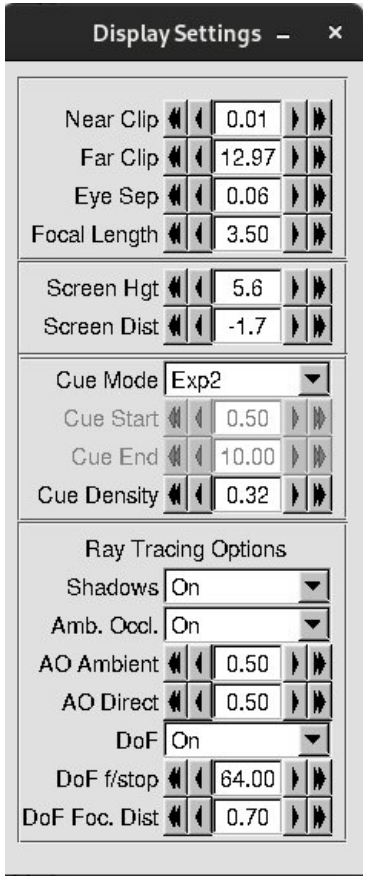


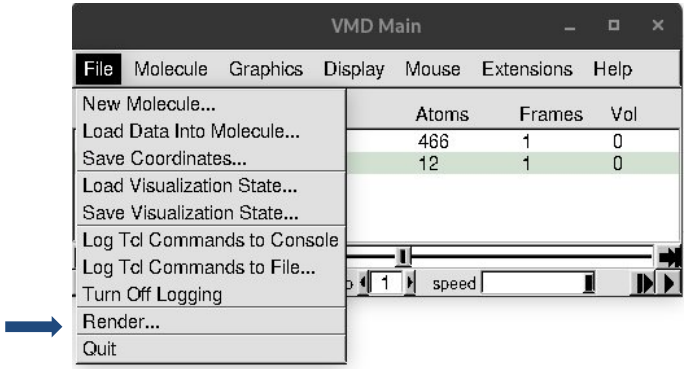
# Rendering: Publication Quality Images

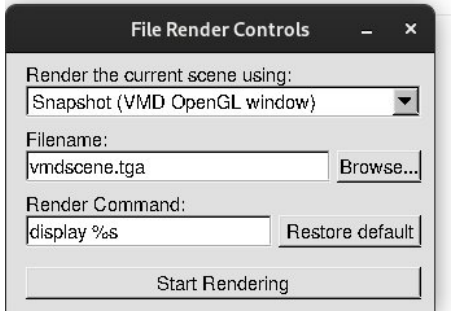
## Go to Display Settings

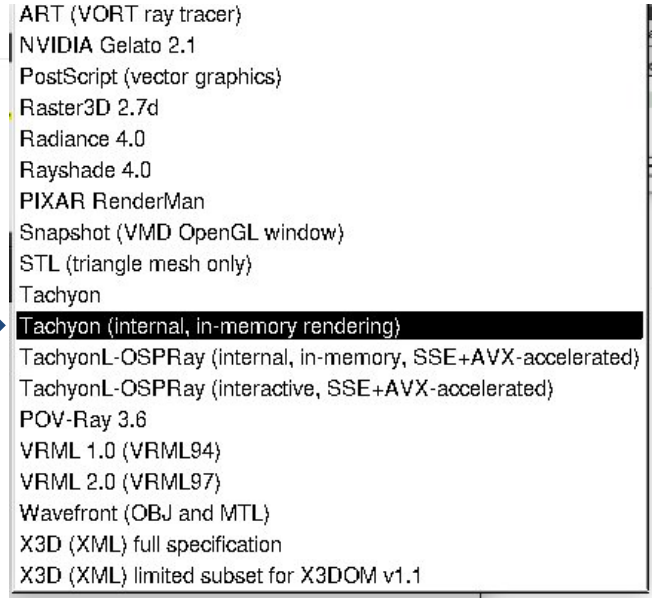


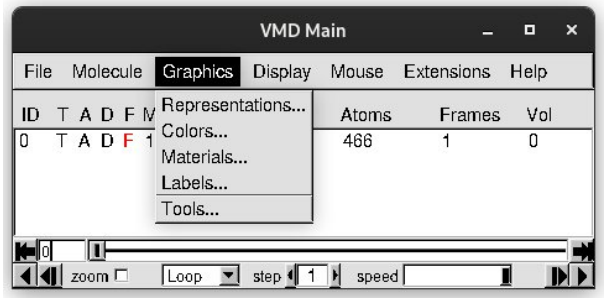

## Go to render settings



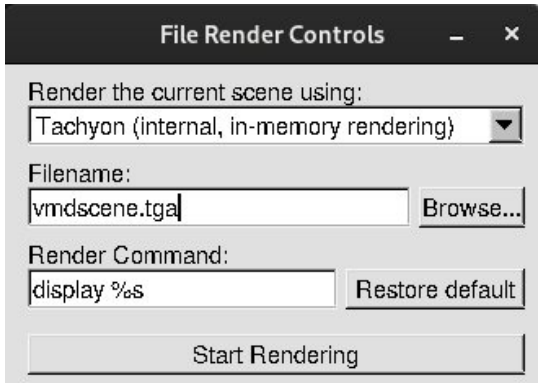



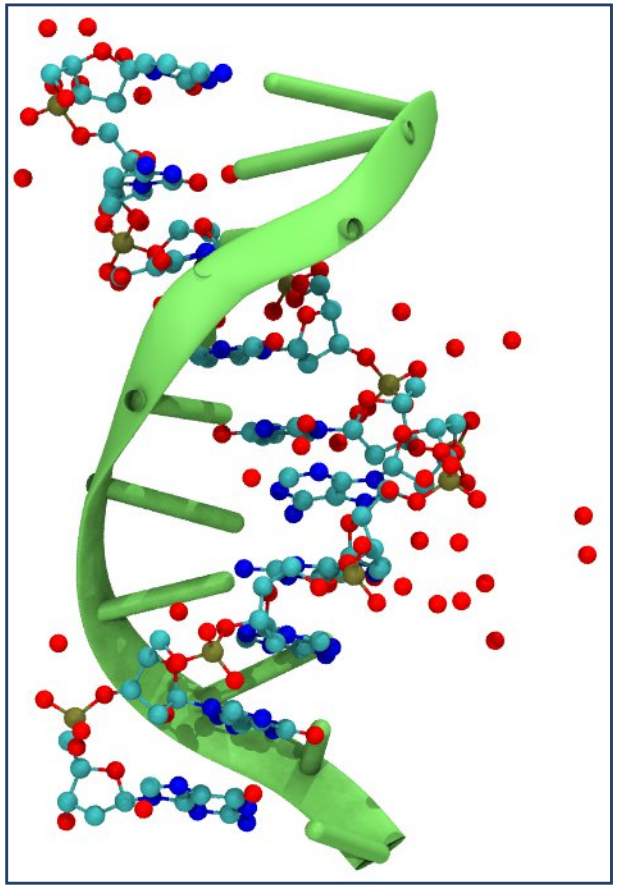
# Rendering: Publication Quality Images

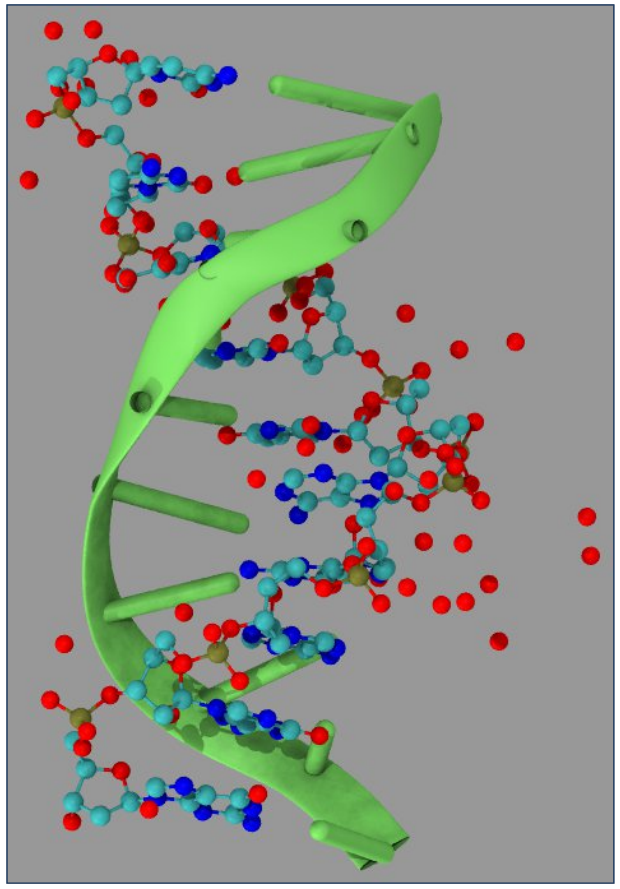



color > background > grey

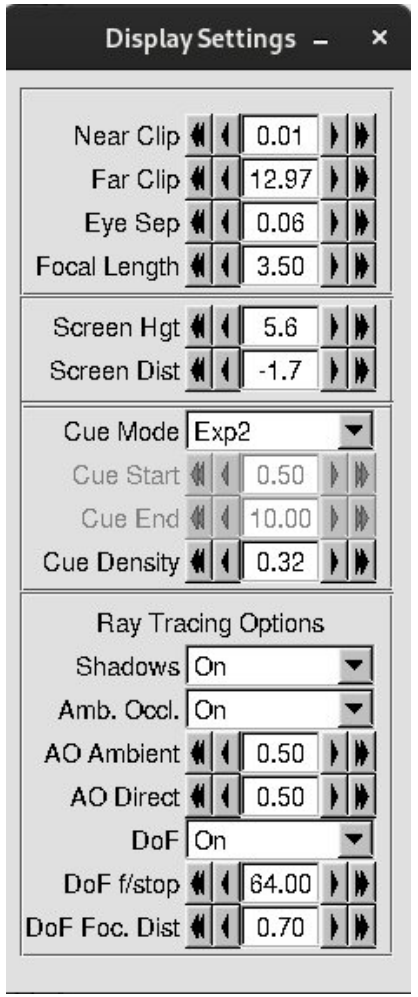
## Render Image



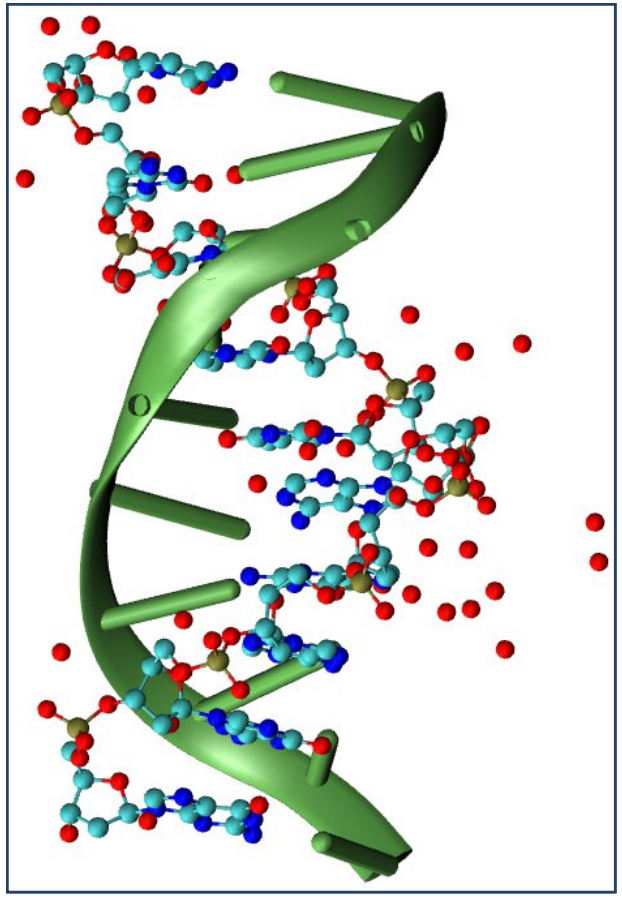


## Fine tune Settings

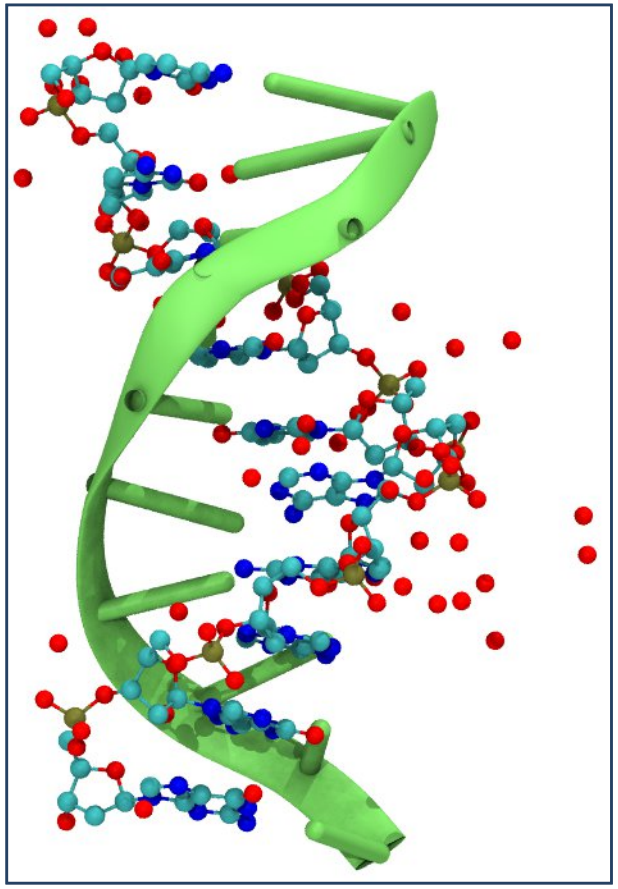


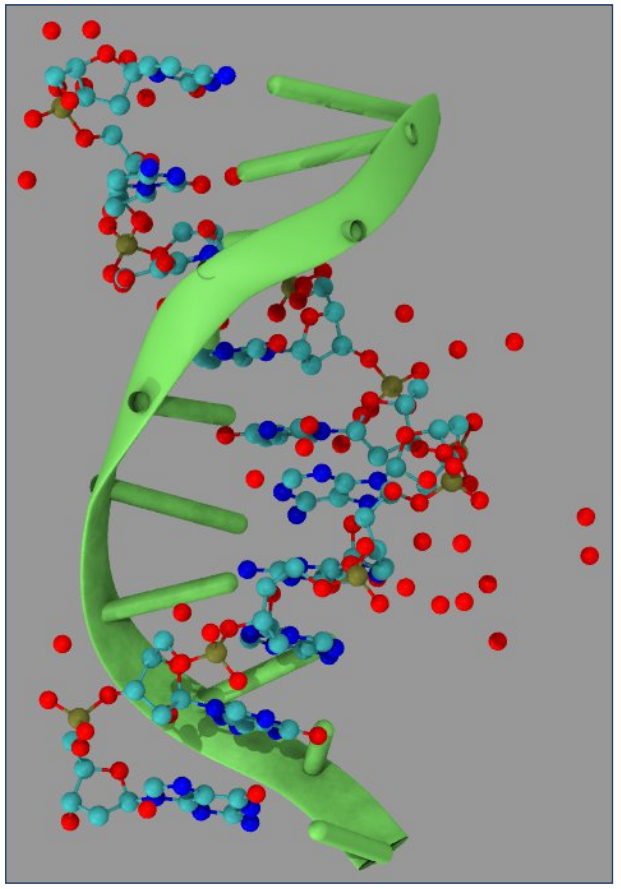
# Rendering: Publication Quality Images

Using VMD Snapshot



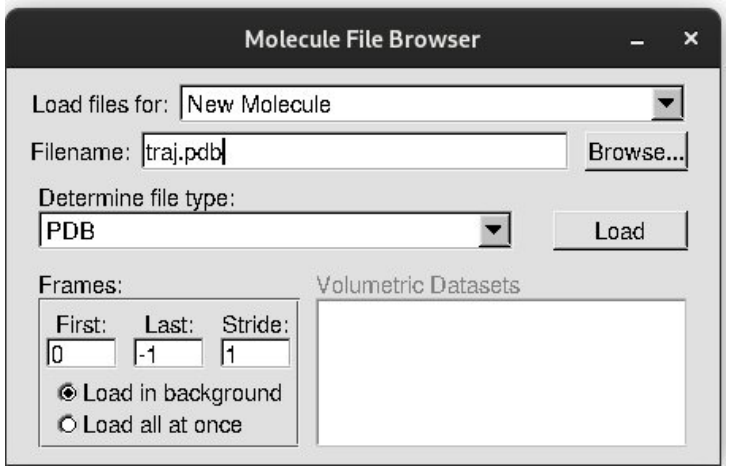
Using Tachyon Internal

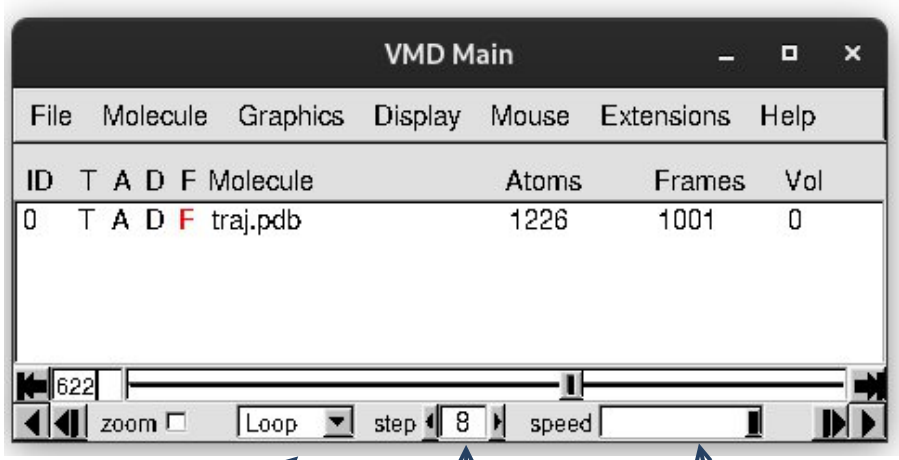




# VMD: Loading Trajectories

Load trajectory



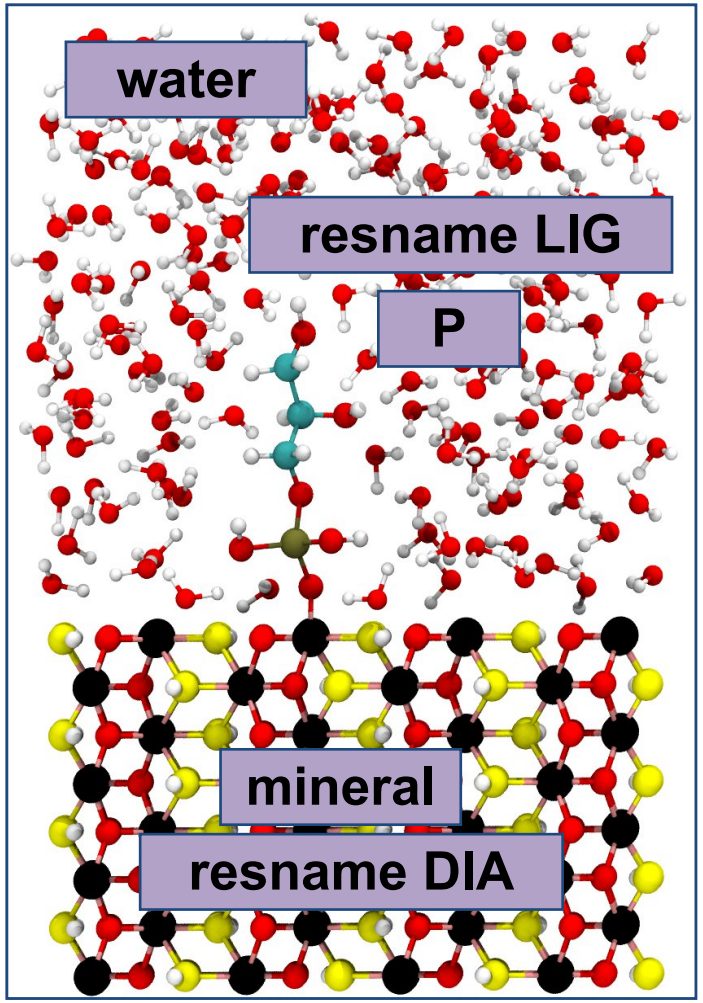




Load custom Frames

Loop/rock

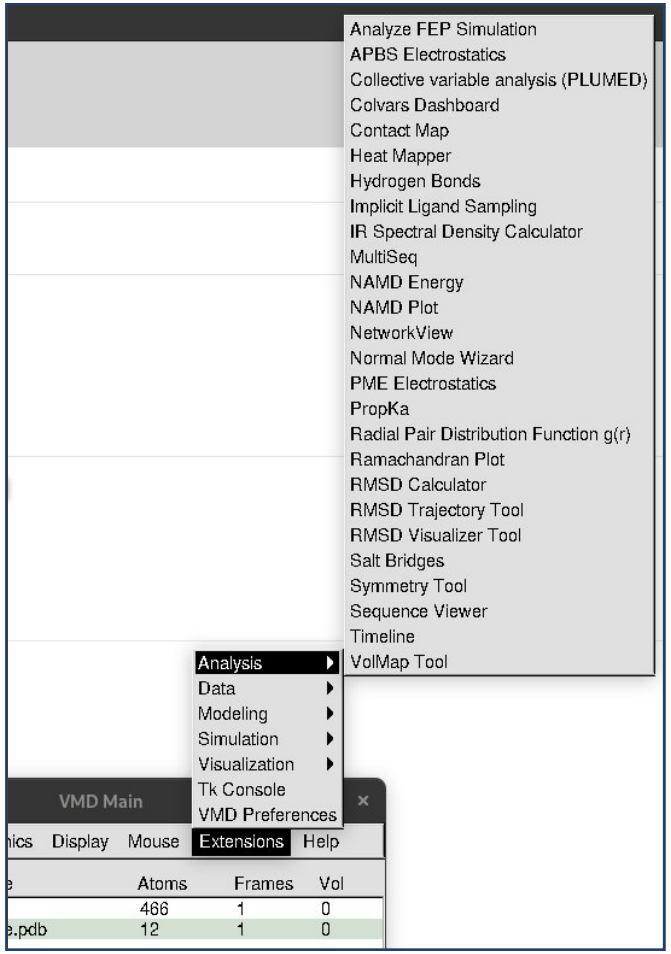
show every 8th step

Speed of frames




/course_data/BioPhyCourse/2026/02_Visualisation_Tools/04_VMD/data/

# VMD: Trajectory Analysis



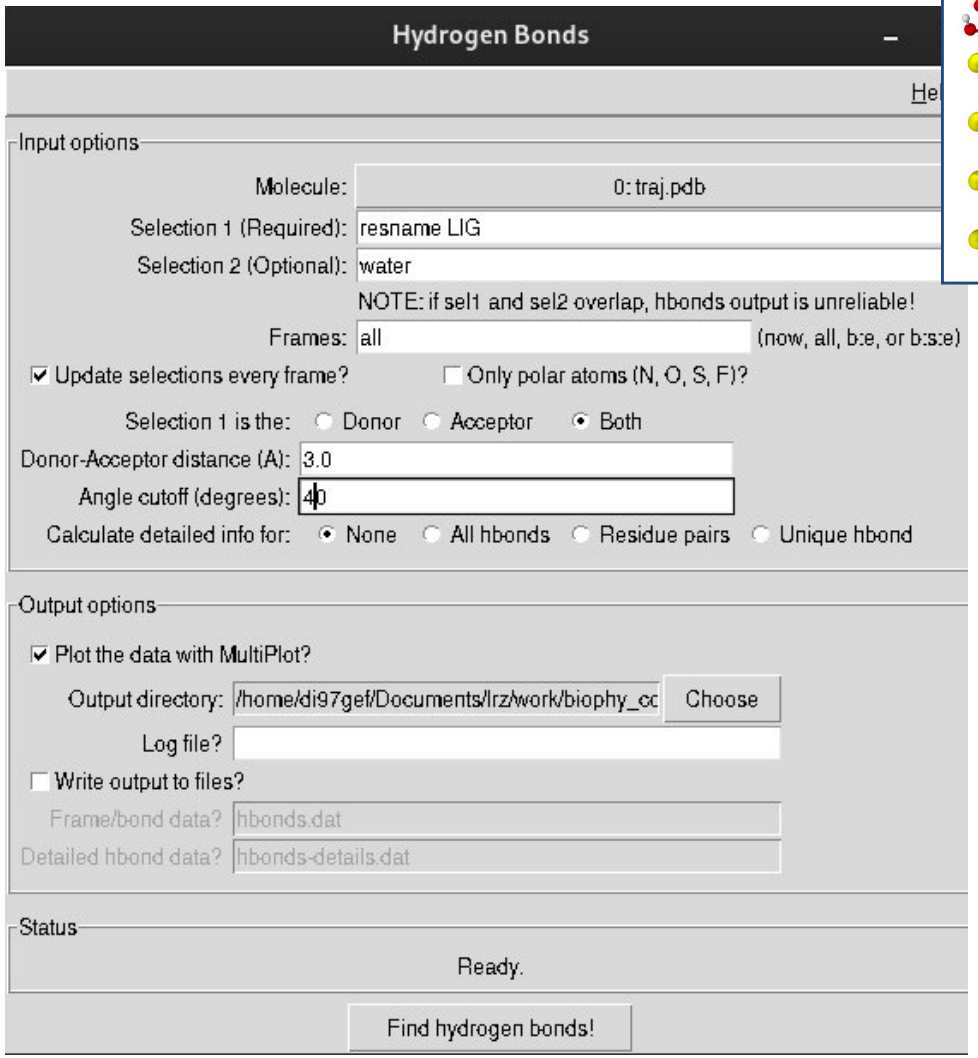

Hydrogen Bond Analysis

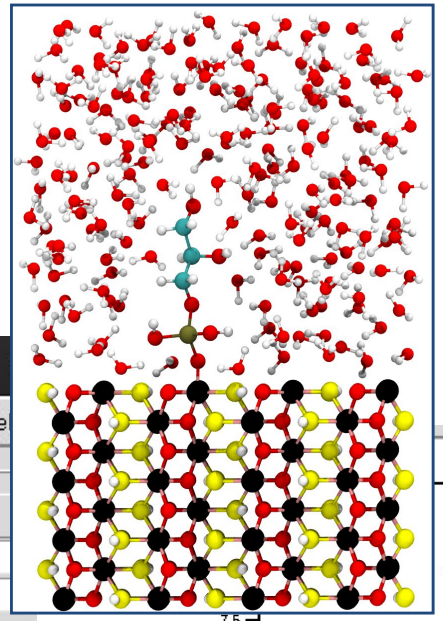
Radial Pair Distribution Function

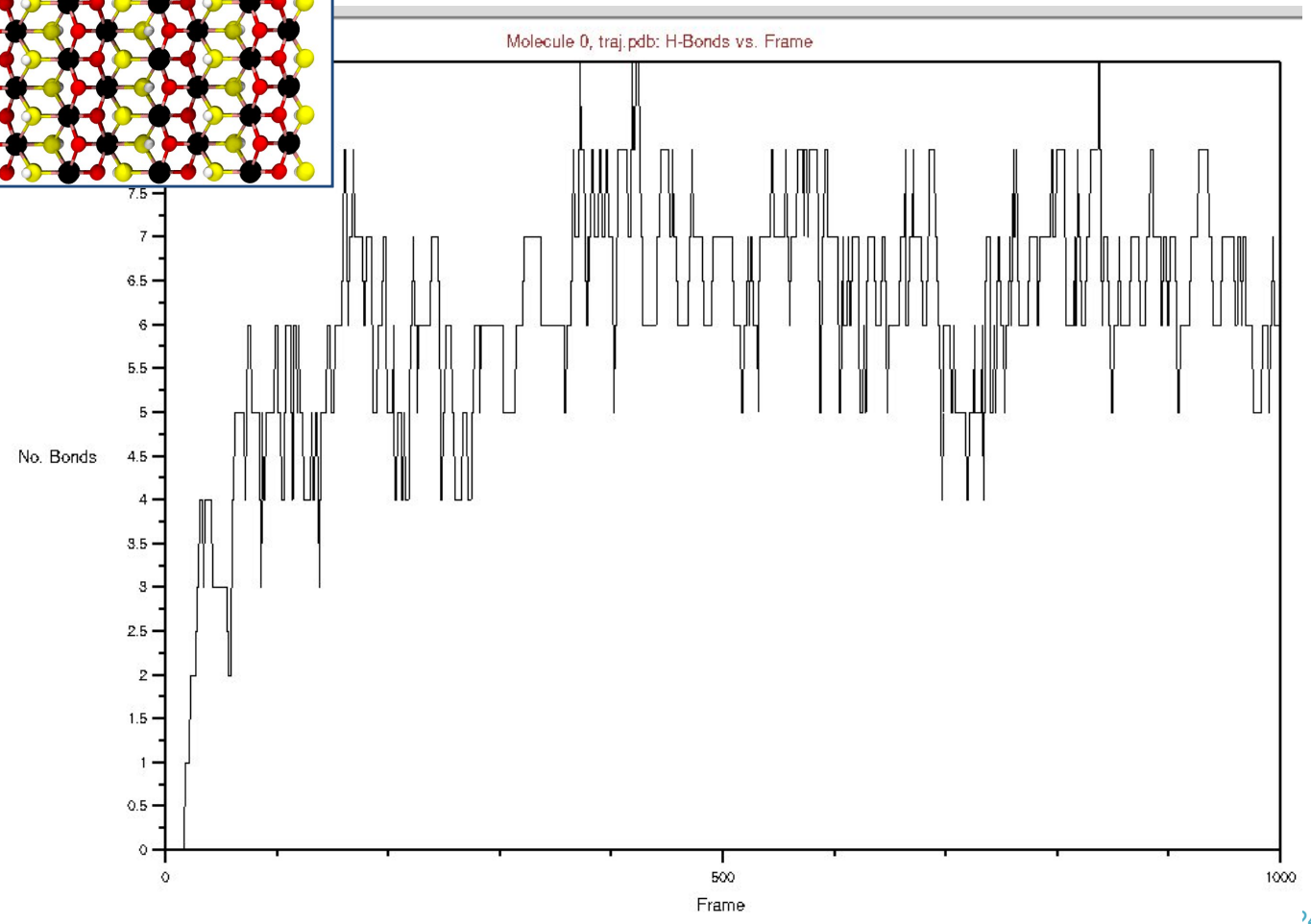
RMSD

# VMD: Trajectory Analysis

Hydrogen Bond Analysis

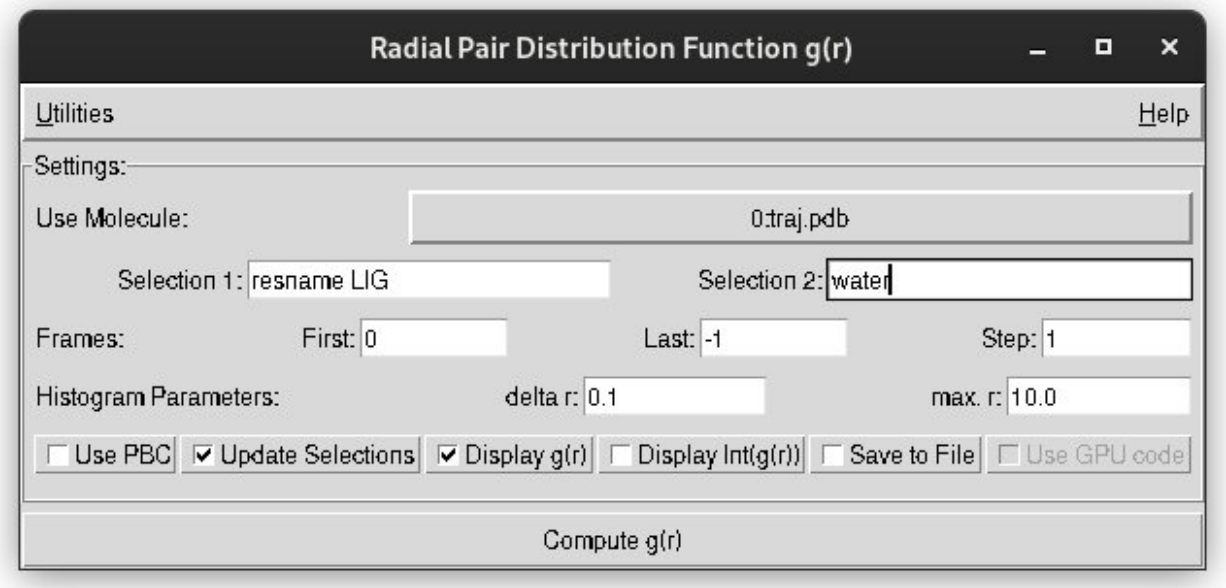


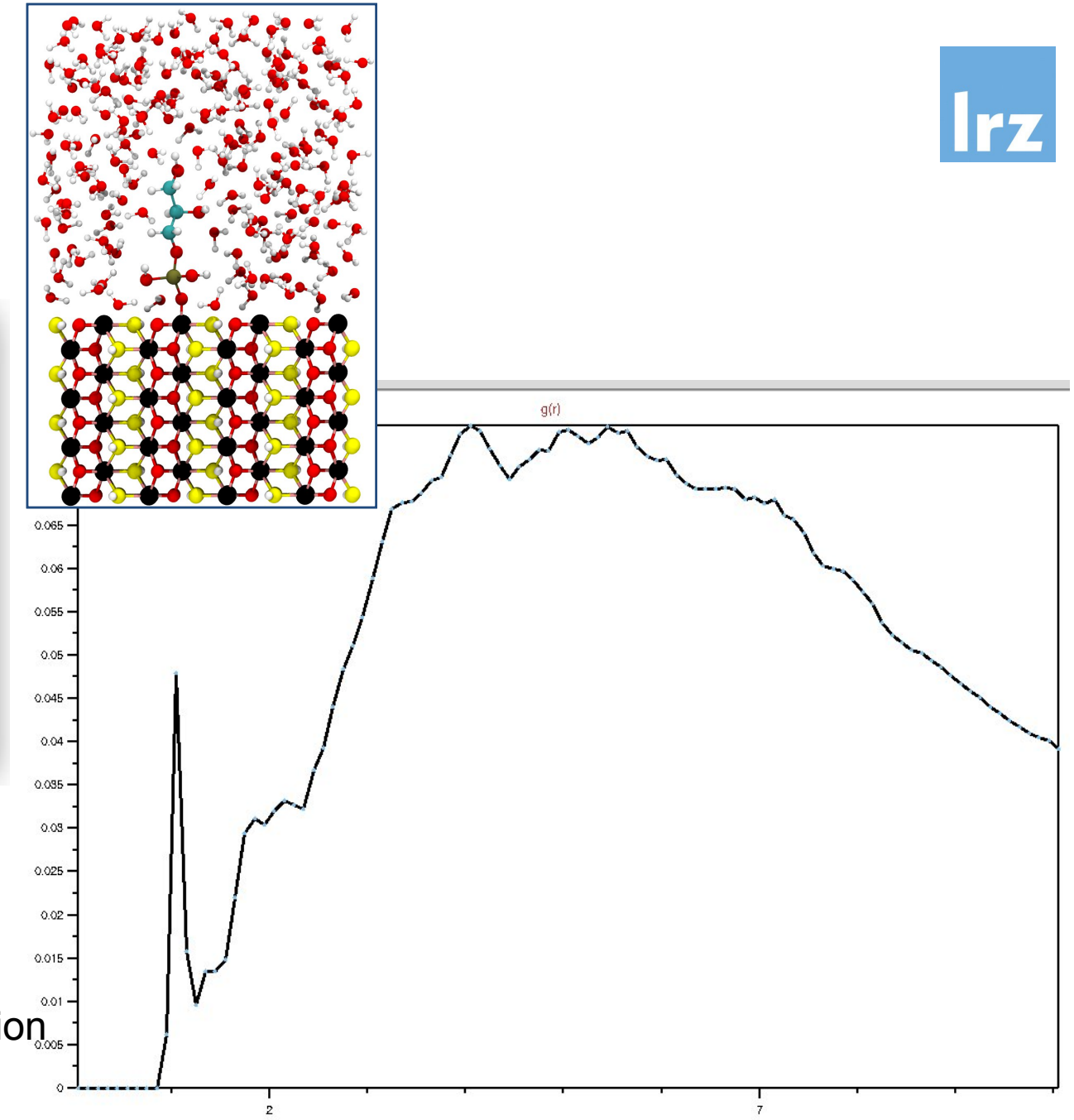





# VMD: Trajectory Analysis

Radial Pair Distribution Function

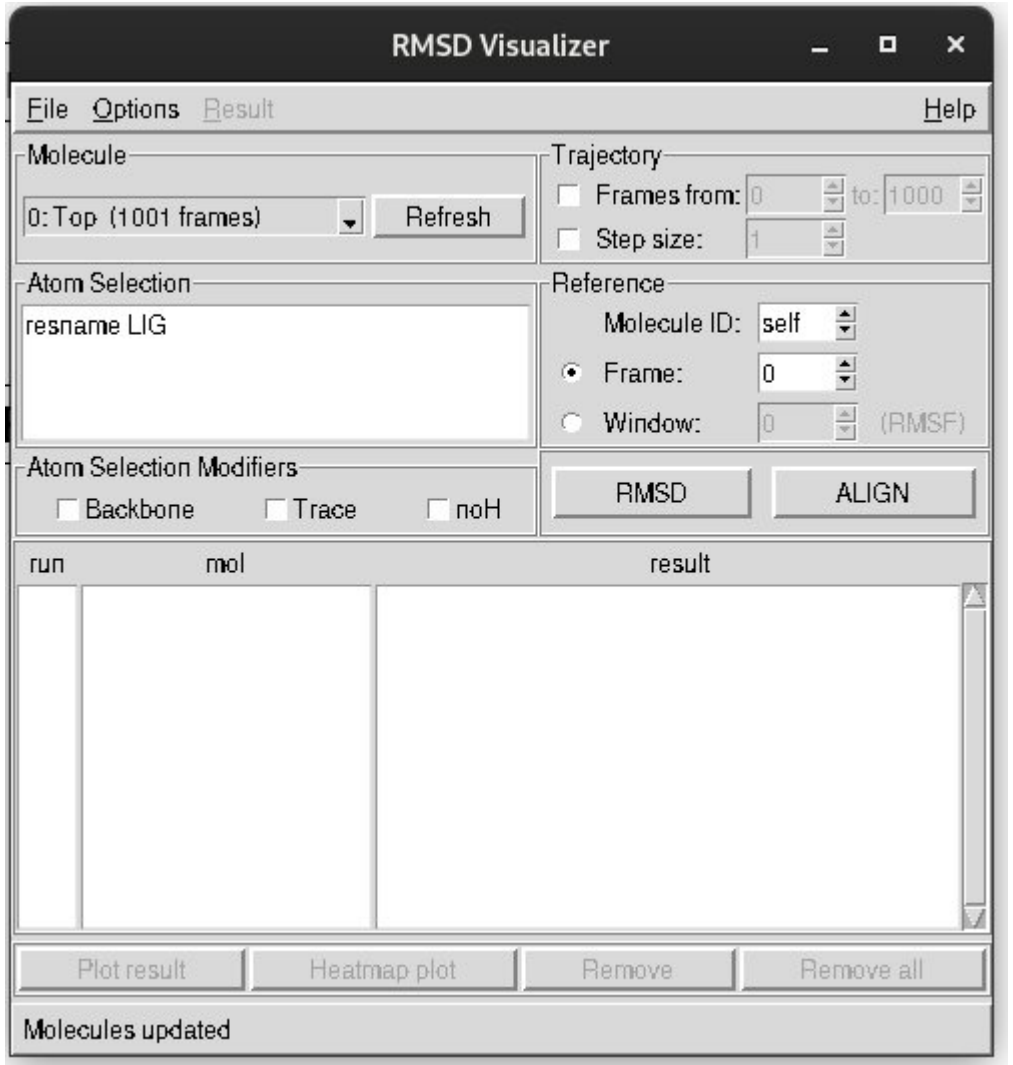


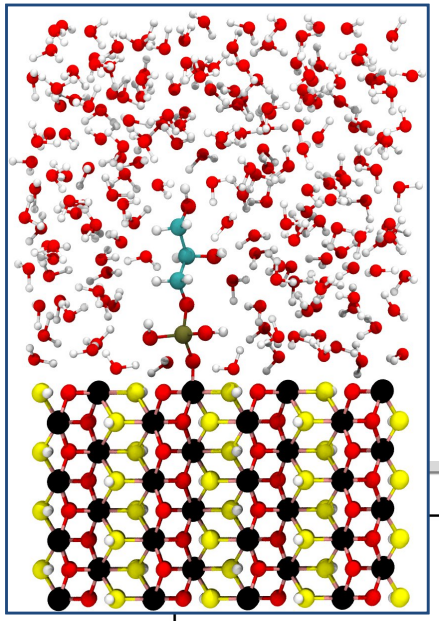



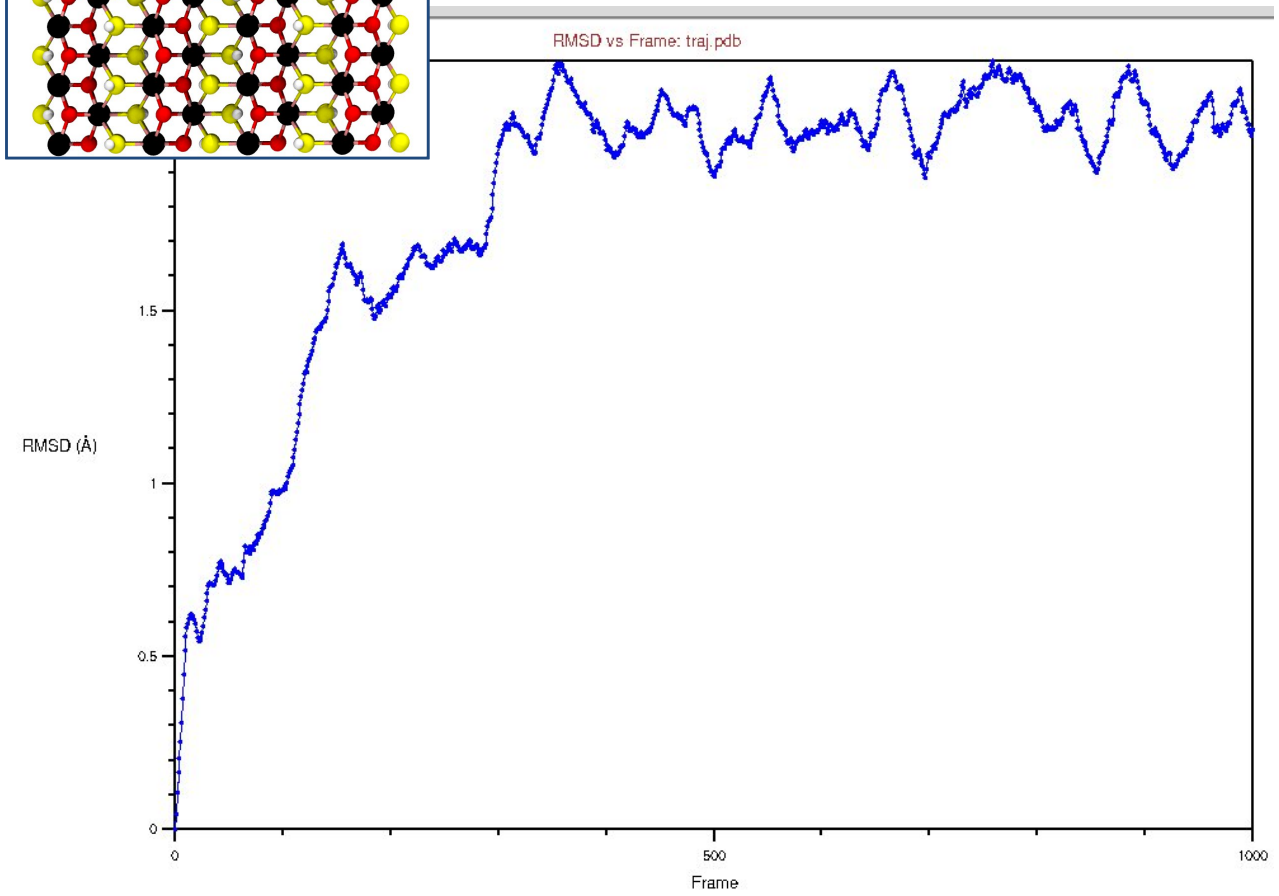

Extensions > Analysis > Radial Pair Distribution Function

# VMD: Trajectory Analysis

RMSD Visualizer Tool



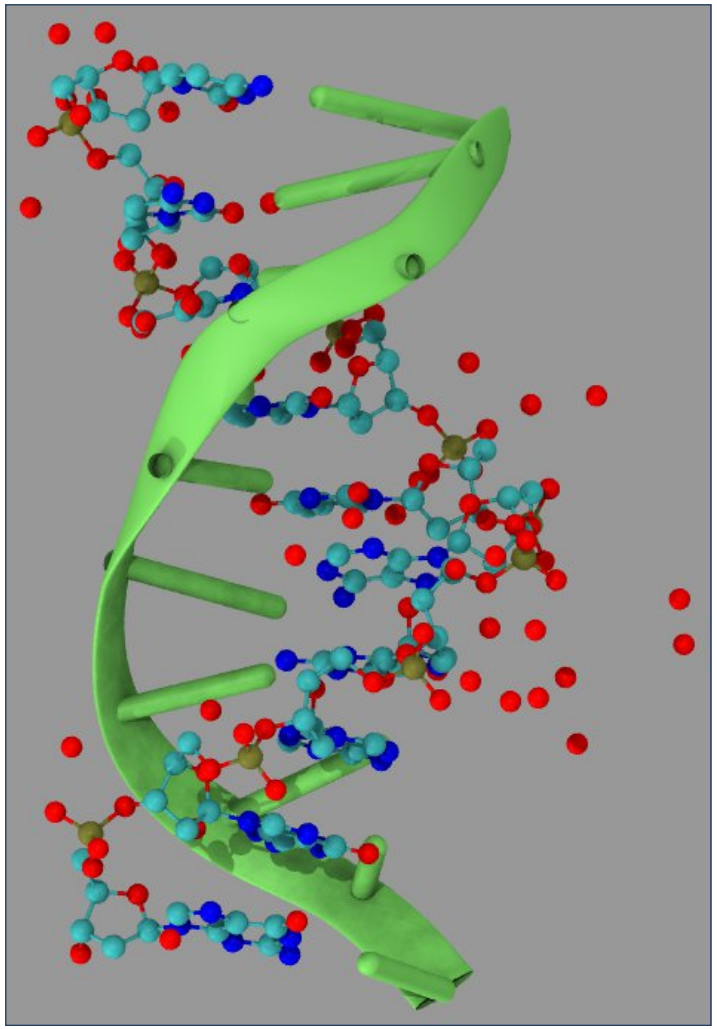


# VMD: Summary

- Visualizes molecular structures (proteins, DNA, membranes)
- Displays MD trajectories
- Supports multiple formats: **PDB, PSF, DCD, GRO, XTC, etc**
- Advanced rendering: cartoon, licorice, surface, VDW
- Plugins for analysis (RMSD, hydrogen bonds)

Advanced:

- Supports **Tcl and Python scripting** for automation
- http://www.ks.uiuc.edu/Training/Tutorials/#vmd



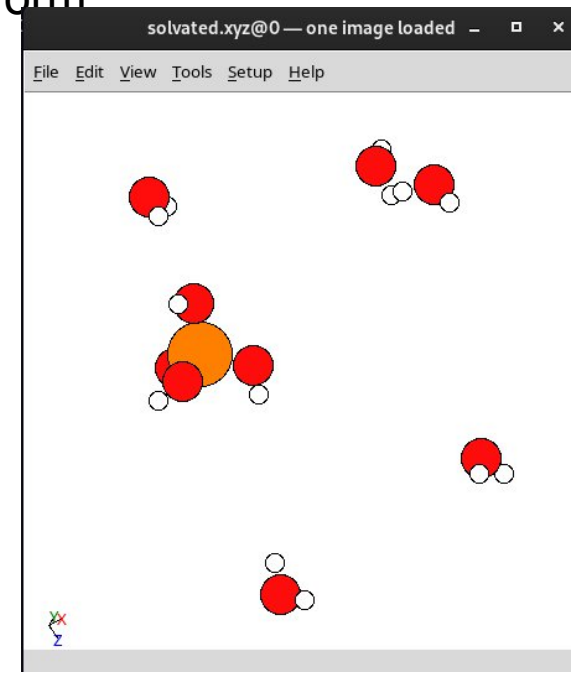
# ATOMIC SIMULATION ENVIRONMENT - ASE

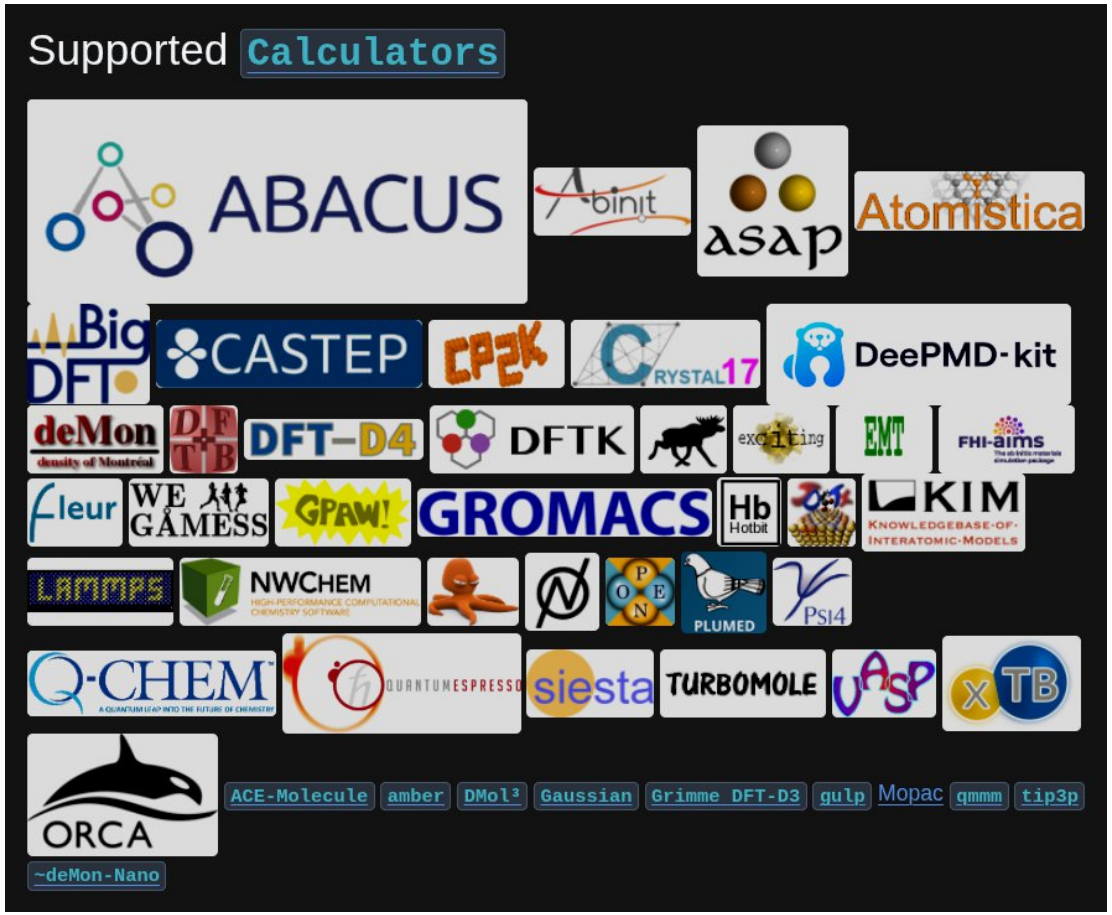
# Introduction to Atomic Simulation Environment (ASE)




- Setting up, steering and analyzing: Python based
- Supports many Calculators/Software
- Easy to use (GUI, CLI)
- Open-source and cross-platform
- Data Analysis tools
- Movie making and Images

```
$ ase --help
$ ase sub-command --help
$ python -m module --help
```




J. Phys.: Condens. Matter 29 273002 (2017)

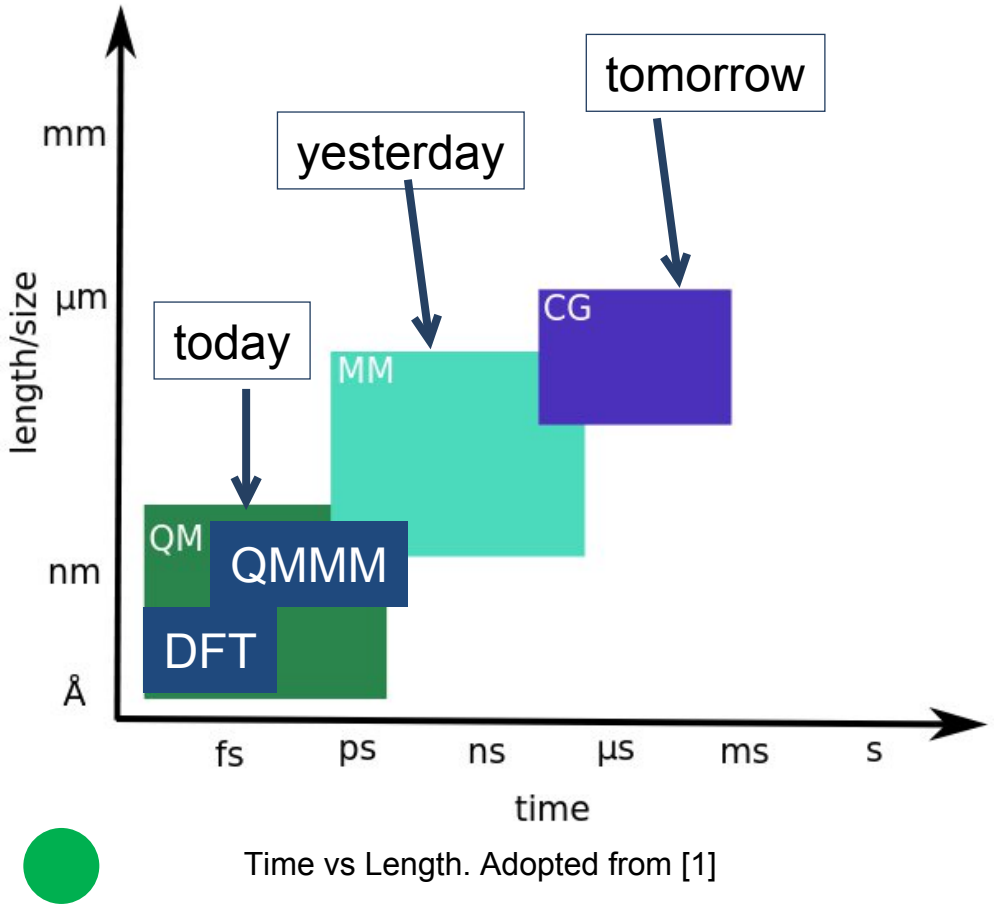
Go to folder:
/course_data/BioPhyCourse/2026/04_Quantum_ESPRESSO_and_CP2K/02_CP2K/data/01-Intro-ASE/

Open file:
MMS_ase_intro.ipynb

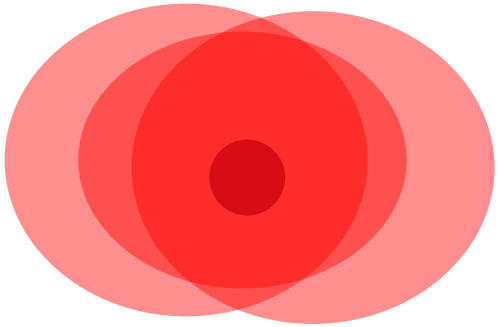
# DENSITY FUNCTIONAL THEORY (DFT)

# Time vs Length Scale

- hierarchy of theories available
- with accuracy computational cost increases
- limits system size and simulation duration
- a wide range of approximations




Time vs Length. Adopted from [1]



QM: Nuclei + Electrons



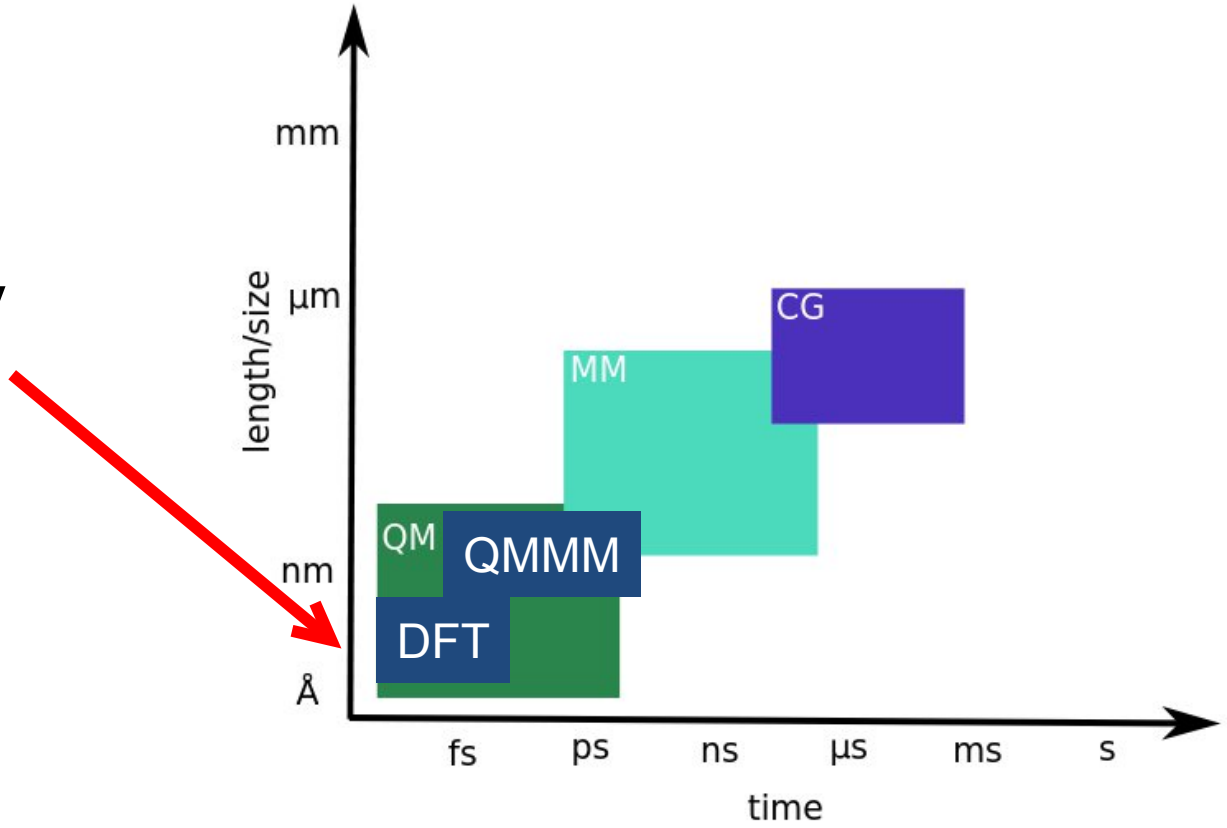
MM: Point Charge



CG: Group of atoms - bead

[1] Q .Rev. Biophys. 49,e4. doi:10.1017/S0033583515000256

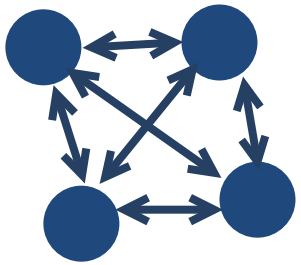
# DFT: Density Functional Theory




Time vs Length. Adopted from [1]

[1] Q .Rev. Biophys. 49,e4. doi:10.1017/S0033583515000256

# Density Functional Theory



DFT

- ground state electron density( $\rho(\mathbf{r})$ ) alone determines the system behavior
- overcomes many-body problem

CP2K - Quickstep Method - DFT GPW [1]

- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$\mathrm{E}[\rho(\mathbf{r})] = \mathrm{E}^{\mathrm{T}}[\chi(\mathbf{r})] + \mathrm{E}^{\mathrm{ne}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{ee}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{nn}} + \mathrm{E}^{\mathrm{XC}}[\rho(\mathbf{r})]$$

| K.E electrons | nuclear-electron | electron-electron | nuclear-nuclear | XC |
|---|---|---|---|---|
| Gaussian | Gaussian+PW | PW | PW | PW |

[1] J. Chem. Phys. 152, 194103 (2020)

# CP2K: the swiss army knife of molecular simulation



- A wide variety of models Hamiltonians
  - classical
  - semi-empirical
  - local and non-local DFT
  - MP2 & RPA
  - Combinations (e.g. QM/MM)
- Various algorithms
  - Molecular dynamics & Monte Carlo
    - NVE, NVT, NPT
  - Free energy and PES tools
  - Ehrenfest MD
- Properties
  - Vibrational
  - NMR, EPR, XAS, TDDFT
- Open source & rapid development
  - 1.000.000 lines of code

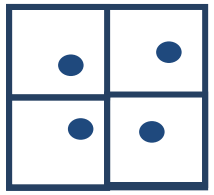
Joost VandeVondele Nanoscale Simulations, ETH Zurich

# CP2K InputFile

$$E[\rho(\mathbf{r})] = E^{T}[\chi(\mathbf{r})] + E^{ne}[\rho(\mathbf{r})] + E^{ee}[\rho(\mathbf{r})] + E^{nn} + E^{XC}[\rho(\mathbf{r})]$$

| K.E electrons | nuclear-electron | electron-electron | nuclear-nuclear | XC |
|---|---|---|---|---|
| Gaussian | Gaussian+PW | PW | PW | PW |

```
&FORCE_EVAL
  &DFT
    CHARGE 0
    &MGRID
      NGRIDS 4
      REL_CUTOFF 30
    &END MGRID
    &QS
      METHOD GPW
      EPS_DEFAULT 1.0E-12
      EXTRAPOLATION ASPC
    &END QS
    &SCF
      SCF_GUESS RESTART
      EPS_SCF 1.0E-8
      CHOLESKY INVERSE
      &DIAGONALIZATION
        ALGORITHM STANDARD
      &END DIAGONALIZATION
      &MIXING
        METHOD  BROYDEN_MIXING
        NBROYDEN 16
      &END MIXING
    &END SCF
```



Grids/Mesh

QM Method and Settings

Fine tune until consistent solution is reached

```
    &XC
      &XC_FUNCTIONAL PBE
      &END XC_FUNCTIONAL
      &VDW_POTENTIAL
            POTENTIAL_TYPE PAIR_POTENTIAL
            &PAIR_POTENTIAL
              PARAMETER_FILE_NAME dftd3.dat
              TYPE DFTD3
              CALCULATE_C9_TERM .TRUE.
              REFERENCE_C9_TERM .TRUE.
              REFERENCE_FUNCTIONAL PBE
            &END PAIR_POTENTIAL
      &END VDW_POTENTIAL
    &END XC
  &END DFT
&END FORCE_EVAL
'''
```

VdW and XC

```
&CELL
   PERIODIC XYZ
   A 17.8    0.      0.
   B 0.     19.15    0.
   C 0.      0.      23.21
&END CELL
```

cell dimensions

```
&KIND O
   BASIS_SET SZV-MOLOPT-GTH
   POTENTIAL GTH-PBE
&END KIND
&KIND H
   BASIS_SET SZV-MOLOPT-GTH
   POTENTIAL GTH-PBE
&END KIND
```

Basis Sets and N

```
calc = CP2K(label='benzene',
            cutoff=250,
            basis_set='SZV-MOLOPT-GTH',
            pseudo_potential='auto',
            charge=0,
            xc='PBE',
            print_level='LOW',
            max_scf=20,
            inp=inp)
```

-project name
-electronic cutoff
-electronic wavefunction
-nuclear
-total System Charge
-exchange correlation
-output setting
-max number of SCF cycles

# Example problem overview

## Importance of Phosphorus

### Essential for plant and animal life


Part of DNA, RNA


Brain functioning


Oxygen delivery


Photosynthesis [1]


Improve flowering


Root development

### Limiting nutrient for plant growth

- Only small amount of P in soil
- Unlike C, N, O, H: P no gaseous phase
- Forests evolved strategies [2]
- Used in fertilizers: global food security





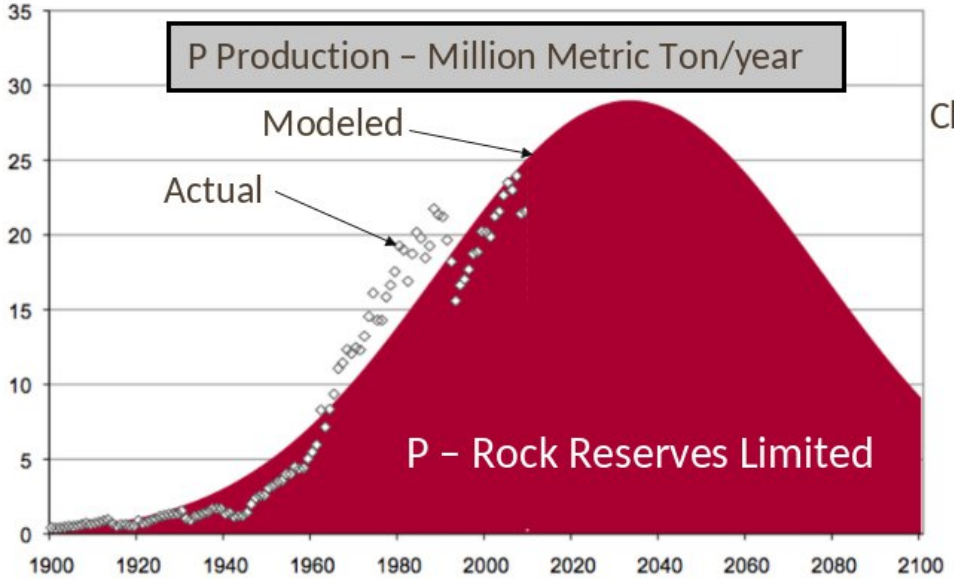


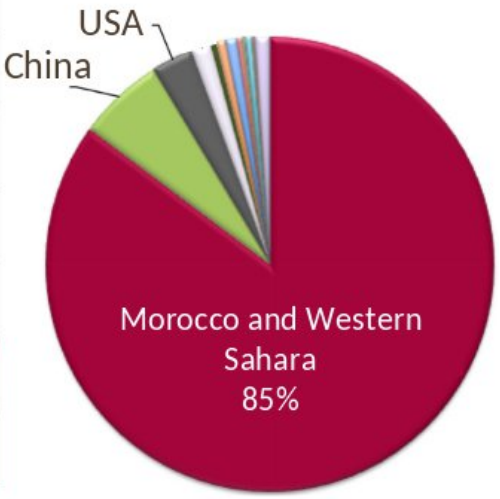
[1] Plants 12(15), 2861 (2023)
[2] Plant and Soil 348, 29 (2011)

# Example problem overview

## P Peak Scenario

- Peak P scenario could happen soon [1]
- Oil Crisis in 1970; NO alternative for P



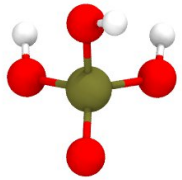

[1] Sustainability, 3(10), 2027-2049 (2011)



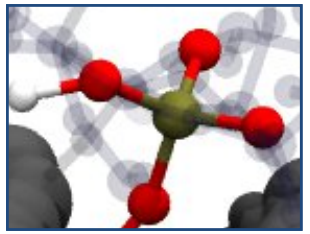
Carrot 17 Pro
1.299 €

IPhoney 200 Pro
6.99 €

# Example problem overview



solvated



Phosphoric acid ($H_3PO_4$)

Phosphate ($HPO_4^{-2}$) [2]

[2] Soil Sci. Plant Anal. 11, 835–842 (1980)

AIM

/course_data/BioPhyCourse/2026/04_Quantum_ESPRESSO_and_CP2K/02_CP2K/02-CP2K-DFT-ASE/

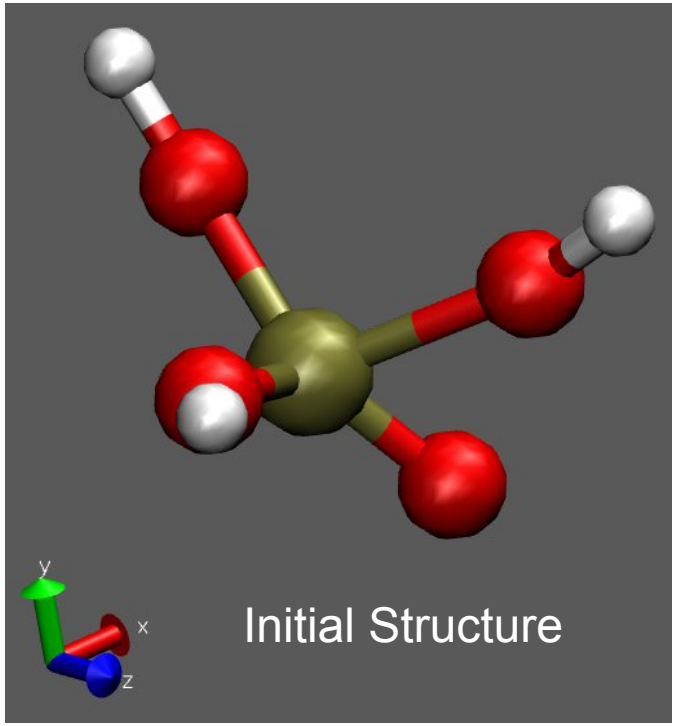
01-CP2K-Intro: Here we optimize Phosphoric acid using CP2K inputfile

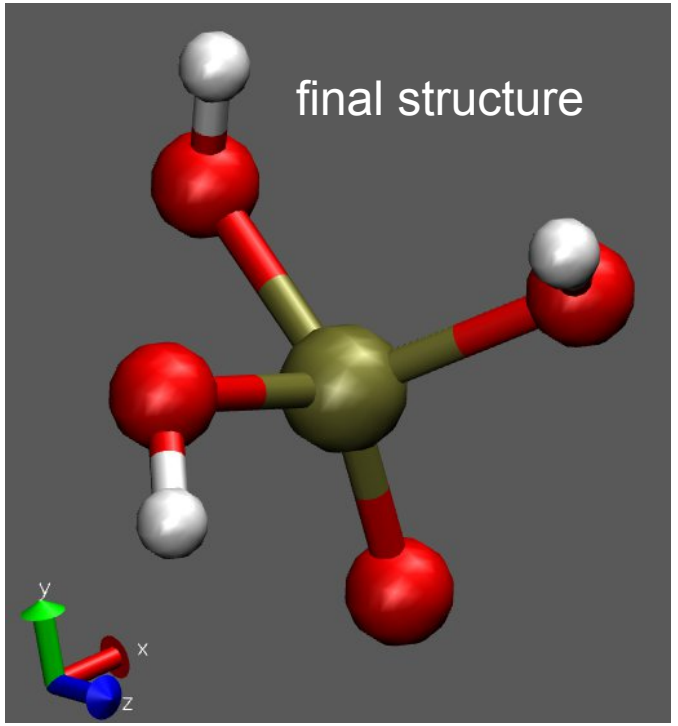
02-CP2K-ASE: Here we solvate Phosphoric acid to see if it deprotonates using ASE+CP2K

# Example problem overview

/course_data/BioPhyCourse/2026/04_Quantum_ESPRESSO_and_CP2K/02_CP2K/02-CP2K-DFT-ASE/01-CP2K-intro

Here we optimize Phosphoric acid using CP2K inputfile



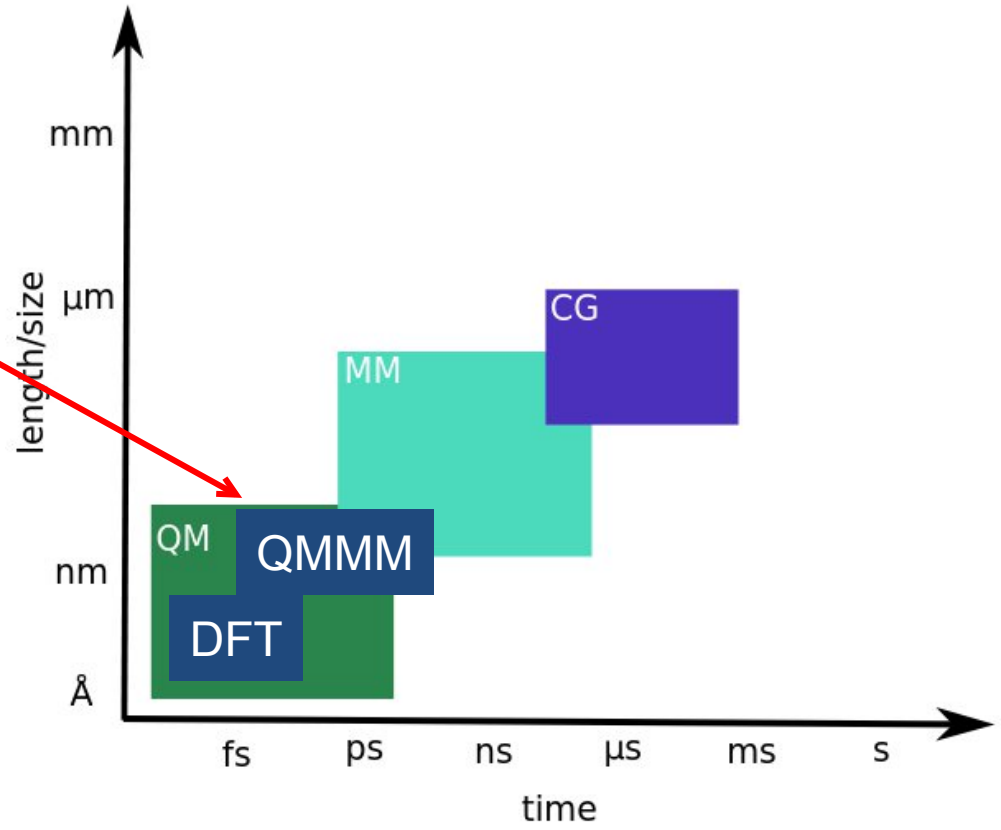



# Example problem overview

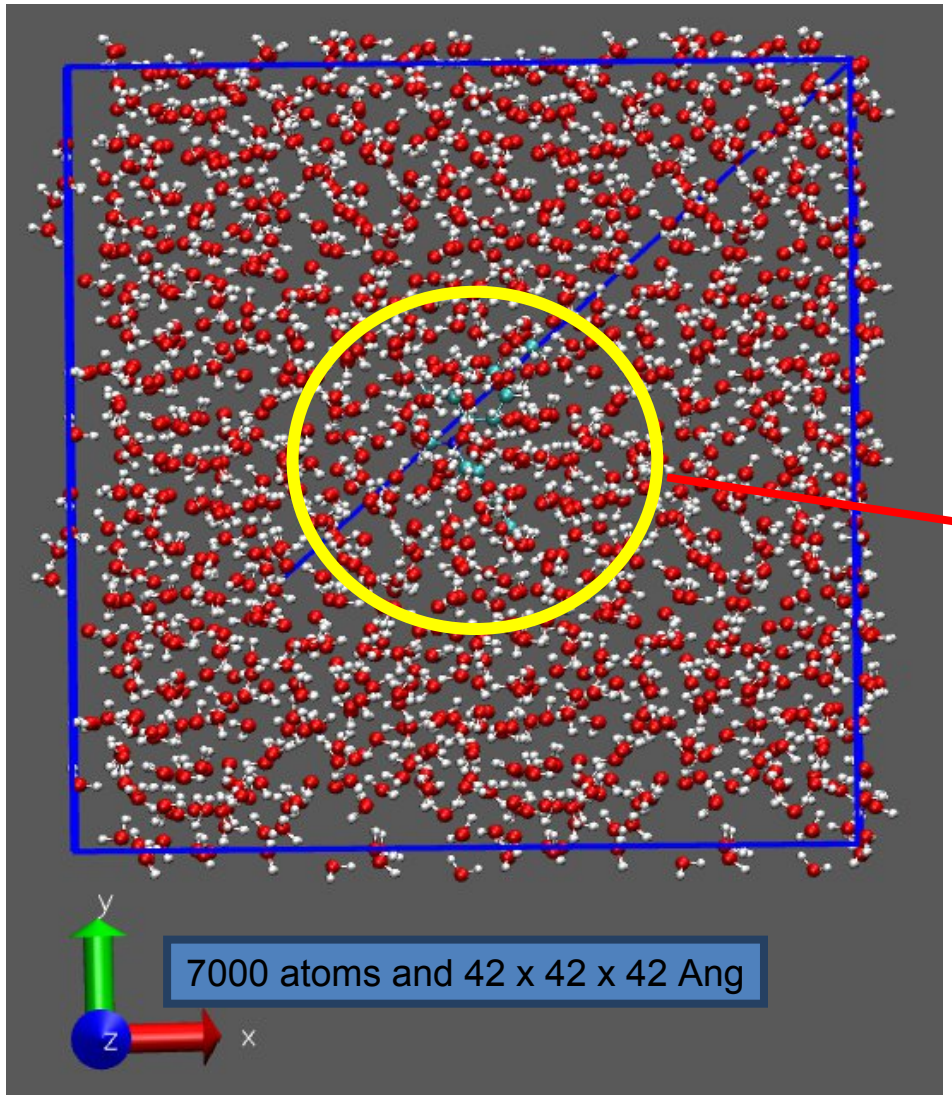
/course_data/BioPhyCourse/2026/04_Quantum_ESPRESSO_and_CP2K/02_CP2K/02-CP2K-DFT-ASE/01-CP2K-intro

Here we solvate Phosphoric acid to see if it deprotonates using ASE+CP2K

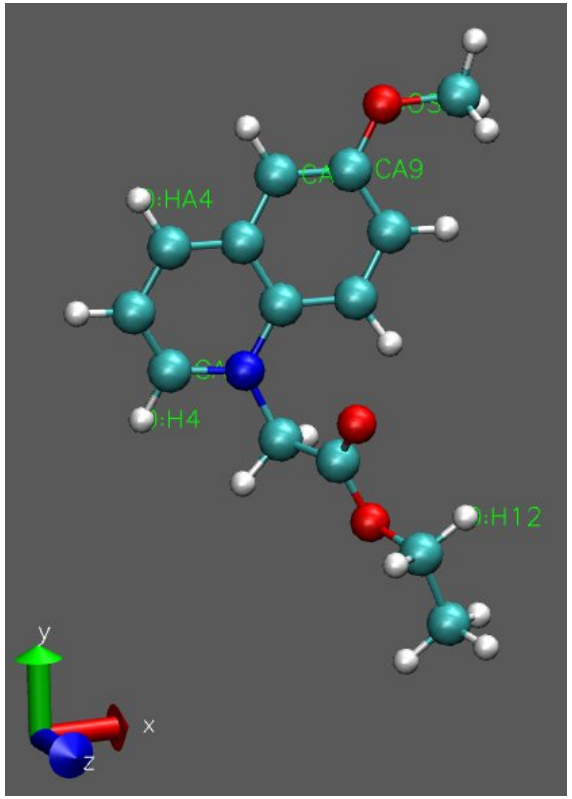
# QM/MM



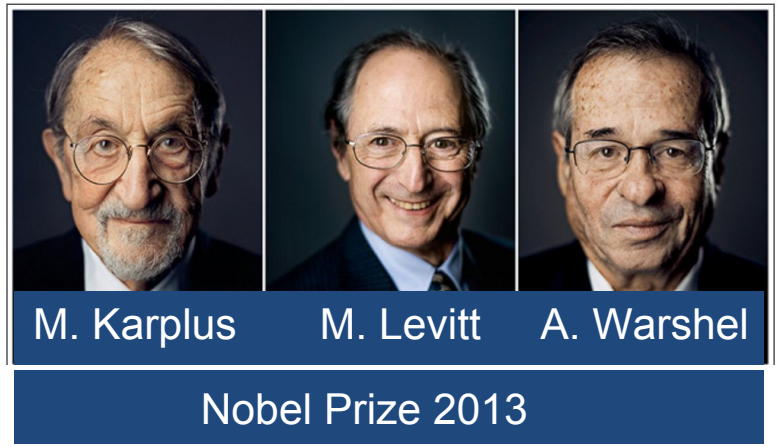
# Why QM/MM



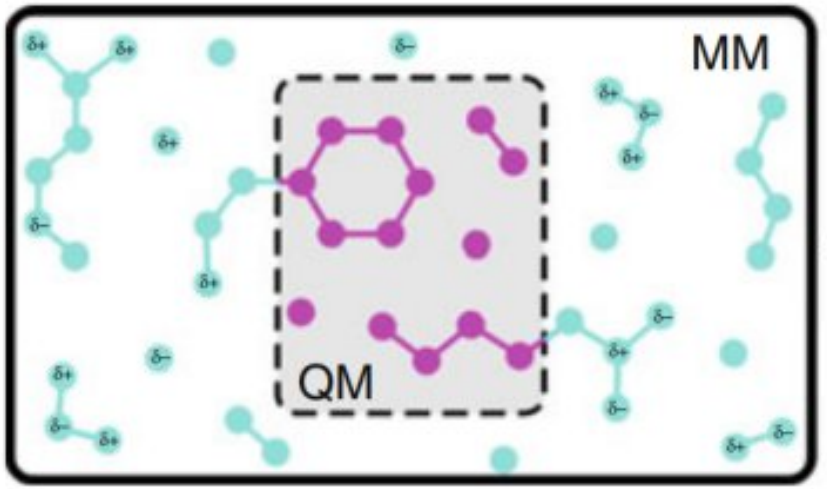

N-(6-methoxyquinolyl) acetoethyl ester



# Why QM/MM



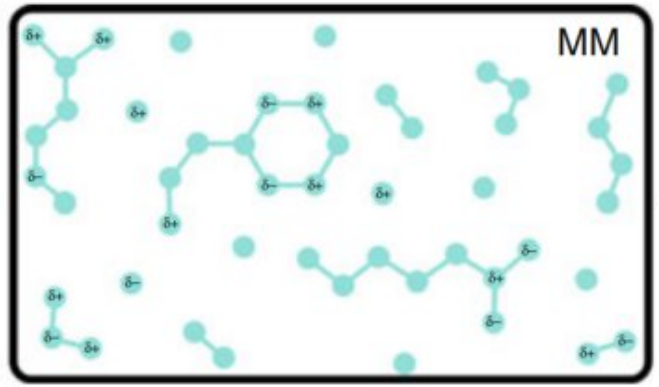

- Pure QM: expensive for 1000+ atoms
- Pure MM: not all reactions defined
- Relevant part with QM: Density Functional Theory [1]
- Rest with low level MM: Mostly All - Atoms FF [2,3]
- QM/MM Coupling [4,5]
- Advantage: Longer & accurate simulations at reduced cost [5]

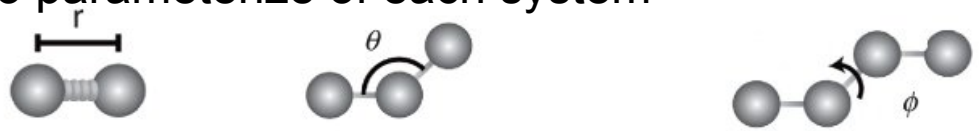



[1] J. Chem. Phys. 152, 194103 (2020)
[2] J. Phys. Chem. B 108, 1255–1266 (2004)
[3] J. Phys. Chem. 91, 6269–6271 (1987)
[4] J. Chem. Theory Comput. 1, 1176–1184 (2005)
[5] Angew. Chem. 48, 1198–1229 (2009)

# Molecular Mechanics (MM)

- Functional forms and parameters define potential energy
- potentials derived from experiments or QM level simulations
- Electrons neglected in the model
- Cannot explain bond-breaking or forming, charge transfer, polarization
- need to parameterize of each system





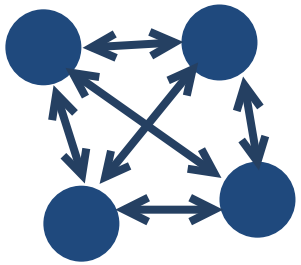


$$E_{system} = \underbrace{\sum k_b(r_{ij} - r_0)^2 + \sum k_\theta(\theta_{ijk} - \theta_0)^2 + \sum k_\phi[1 + \cos(n\phi - \delta)]}_{E_{bonded}} + \underbrace{\sum_{\substack{i,j \\ i \neq j}} e^2 \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \sum_{\substack{i,j \\ i \neq j}} \epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right]}_{E_{nonbonded}}$$

$$\boxed{E_{system} = E_{bonded} + E_{nonbonded}}$$

$$\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)$$
$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$$

# Density Functional Theory (QM)



DFT

- ground state electron density( $\rho(\mathbf{r})$ ) alone determines the system behavior
- overcomes many-body problem

Quickstep Method - DFT GPW [1]

- Uses both Gaussian and Planewaves to speedup calculation
- **Gaussian**: Localized basis set centered on atomic positions – Suitable for molecules
- **Planewave**: Best suited for condensed matter, solids, periodic system

$$\mathrm{E}[\rho(\mathbf{r})] = \mathrm{E}^{\mathrm{T}}[\chi(\mathbf{r})] + \mathrm{E}^{\mathrm{ne}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{ee}}[\rho(\mathbf{r})] + \mathrm{E}^{\mathrm{nn}} + \mathrm{E}^{\mathrm{XC}}[\rho(\mathbf{r})]$$

| K.E electrons | nuclear-electron | electron-electron | nuclear-nuclear | XC |
|---|---|---|---|---|
| Gaussian | Gaussian+PW | PW | PW | PW |

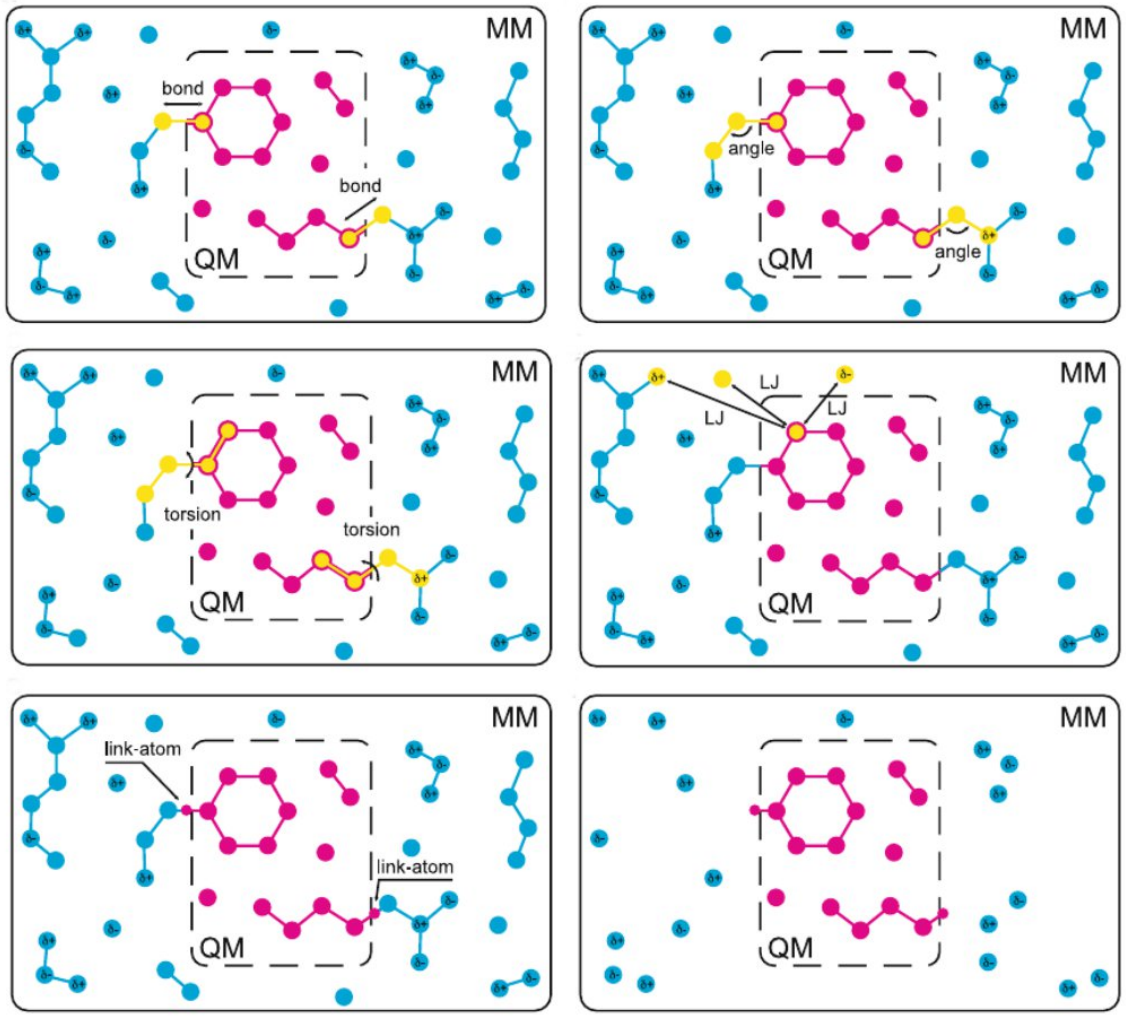
# Additive QM/MM

- $E(\mathbf{r}_\alpha,\mathbf{r}_a)=E^{QM}(\mathbf{r}_\alpha)+E^{MM}(\mathbf{r}_a)+E^{QM/MM}(\mathbf{r}_\alpha,\mathbf{r}_a)$

- Mechanical embedding: Electronic wavefunction isolated
  - QM-MM interactions defined at MM level

- Electrostatic Embedding: MM charges as one electron operators
  - MM polarize QM, but not vice versa

- Polarization Embedding: QM ⇔ MM polarization
  - Computationally expensive



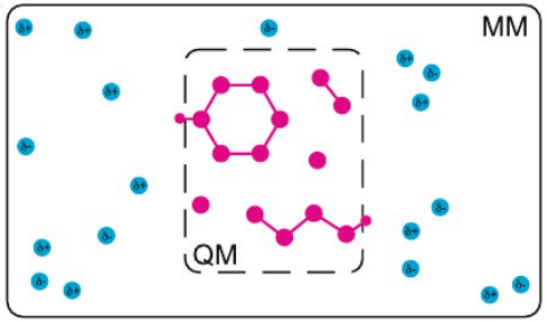

Gerrit Groenhof. Introduction to QM/MM Simulations, Humana Press, Totowa, NJ, 2013.

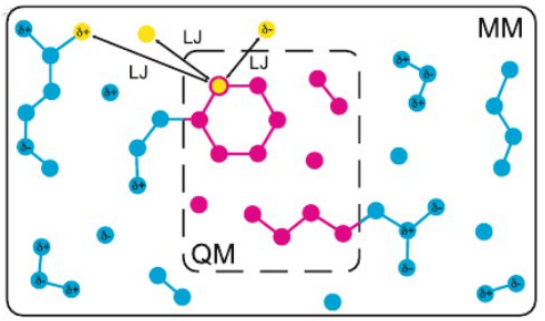
# Electrostatic Embedding

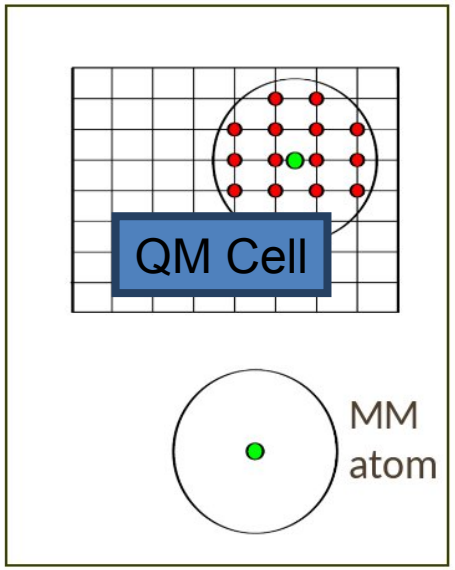
charge density
electron + nuclei

- $$\mathrm{E}_{\mathrm{NB}}^{\mathrm{QM/MM}}(\mathbf{r}_\mathrm{i}, \mathbf{R}_\mathrm{I}) = \frac{1}{4\pi\epsilon_0} \sum_{\mathrm{i}\in\mathrm{MM}} q_\mathrm{i} \int \frac{\rho}{|\mathbf{r}-\mathbf{r}_\mathrm{i}|}\mathrm{d}\mathbf{r} + \sum_{\substack{\mathrm{i}\in\mathrm{MM}\\ \mathrm{I}\in\mathrm{QM}}} \nu_{\mathrm{vdw}}(\mathbf{r}_\mathrm{i}, \mathbf{R}_\mathrm{I})$$
- MM atoms polarize the QM system
- charges of MM atoms and LJ parameters need to be parameterized
- electron-spill: electrons of QM atom close to MM atom gets trapped
- electrostatic embedding: GEEP
- MM Charge density in Gaussian

$$\frac{1}{(\sqrt{\pi}\ r_{c,a})^3} \exp\left(-\left(\frac{|\mathbf{r}-\mathbf{r}_\mathrm{i}|}{r_{c,a}}\right)^2\right)$$



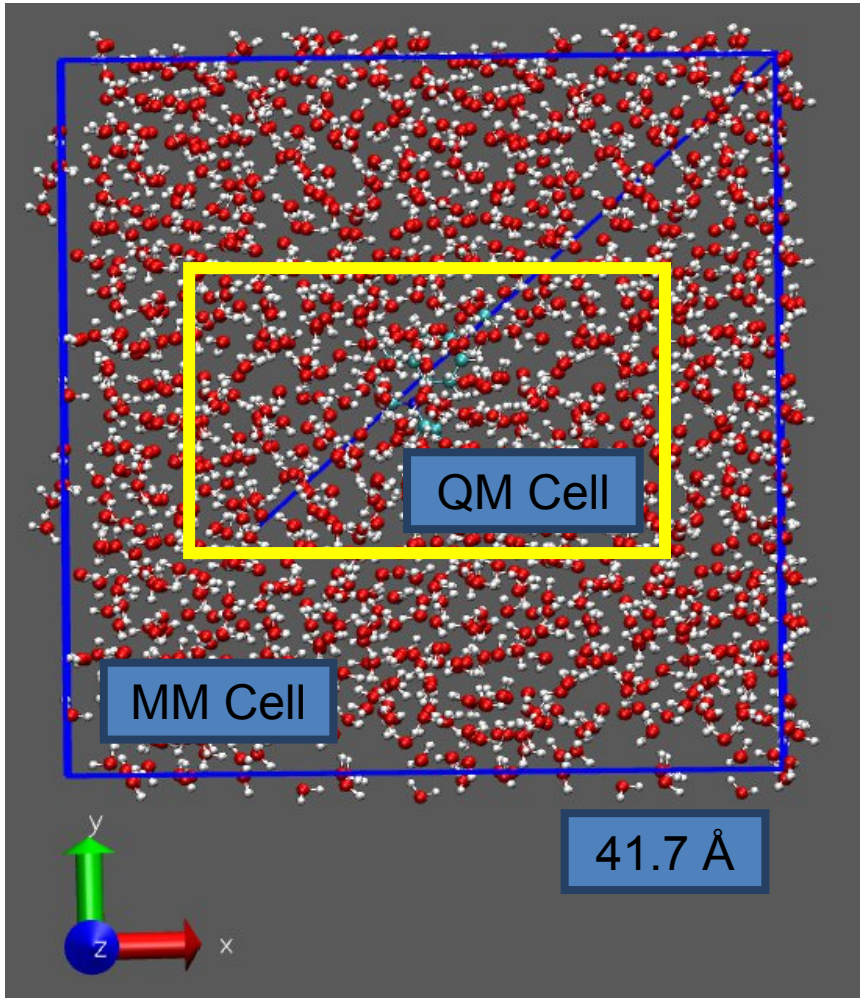



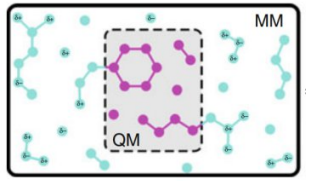




Gerrit Groenhof. Introduction to QM/MM Simulations, Humana Press, Totowa, NJ, 2013.

[GEEP: J. Chem. Th. Comp., 1, pp. 1176-1184 (2005)

# Example1: Overview



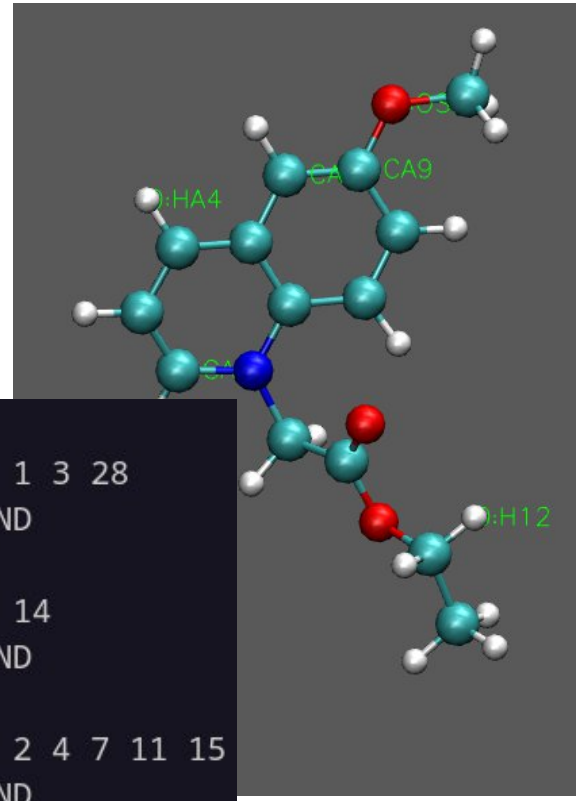

```
&QMMM  ! This defines the QS cell i
  ECOUPL GAUSS
  USE_GEEP_LIB 15
  &CELL
    ABC 13.79565 17.3825 10.76061
    PERIODIC XYZ
  &END CELL
  &PERIODIC  # apply periodic poten
```

```
&SUBSYS
  &CELL
    ABC 41.751 41.751 41.751
    ALPHA_BETA_GAMMA 90 90 90
    PERIODIC XYZ
  &END CELL
```

```
&KIND H
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT H
  POTENTIAL GTH-BLYP-q1
&END KIND
&KIND C
  BASIS_SET DZVP-MOLOPT-GTH
  ELEMENT C
  POTENTIAL GTH-BLYP-q4
&END KIND
```

```
ATOM      8 HC    17.593  24.836  20.763
ATOM      9 HC    19.204  25.485  21.124
ATOM     10 HC    18.086  25.028  22.422
ATOM     11 C32   20.744  21.568  18.721
ATOM     12 H13   21.474  22.379  18.609
ATOM     13 H13   19.956  21.768  18.003
ATOM     14 NA    21.342  20.240  18.435
ATOM     15 CA8   22.670  20.212  17.952
```

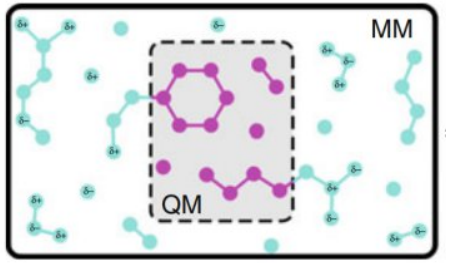
N-(6-methoxyquinolyl) acetoethyl ester




```
&QM_KIND O
  MM_INDEX 1 3 28
&END QM_KIND
&QM_KIND N
  MM_INDEX 14
&END QM_KIND
&QM_KIND C
  MM_INDEX 2 4 7 11 15
&END QM_KIND
&QM_KIND H
```

# QMMM Inputfile

BioPhyCourse-latest/Day2/03_CP2K_QMMM/data/03-CP2K-QMMM/QMMM_MQAE




```
&FORCE_EVAL
  METHOD QMMM
  &DFT
    BASIS_SET_FILE_NAME BASIS_MOLOPT
    CHARGE 1
    MULTIPLICITY 1
    POTENTIAL_FILE_NAME POTENTIAL
    &MGRID
      COMMENSURATE
      CUTOFF 400
    &END MGRID
    &QS
      EPS_DEFAULT 1.0E-12
      METHOD GPW
    &END QS
    &SCF
      EPS_SCF 1.0E-6
      MAX_SCF 300
      &OT T
        MINIMIZER DIIS
        PRECONDITIONER FULL_ALL
        STEPSIZE 1.4999999999999999E-01
      &END OT
    &END SCF
    &XC
      &XC_FUNCTIONAL BLYP
      &END XC_FUNCTIONAL
    &END XC
  &END DFT
  &MM
    &FORCEFIELD
```

DFT Section

```
  &END DFT
  &MM
    &FORCEFIELD
      EI_SCALE14 1.0
      PARMTYPE AMBER
      PARM_FILE_NAME MQAE.prmtop
      VDW_SCALE14 1.0
      &SPLINE
        EMAX_SPLINE 1.0E14
        RCUT_NB [angstrom] 12
      &END SPLINE
    &END FORCEFIELD
    &POISSON
      &EWALD
        ALPHA .40
        EWALD_TYPE SPME
        GMAX 80
      &END EWALD
    &END POISSON
  &END MM
  &QMMM  ! This defines the QS cell in t
    ECOUPL GAUSS
    USE_GEEP_LIB 15
    &CELL
      ABC 13.79565 17.3825 10.76061
      PERIODIC XYZ
    &END CELL
    &PERIODIC  # apply periodic potentia
      #turn on/off coupling/recoupling c
      &MULTIPOLE ON
      &END MULTIPOLE
```

MM Section

```
    &END POISSON
  &END MM
  &QMMM  ! This defines the QS cell i
    ECOUPL GAUSS
    USE_GEEP_LIB 15
    &CELL
      ABC 13.79565 17.3825 10.76061
      PERIODIC XYZ
    &END CELL
    &PERIODIC  # apply periodic poten
      #turn on/off coupling/recouplin
      &MULTIPOLE ON
      &END MULTIPOLE
    &END PERIODIC
    &QM_KIND O
      MM_INDEX 1 3 28
    &END QM_KIND
    &QM_KIND N
      MM_INDEX 14
    &END QM_KIND
    &QM_KIND C
      MM_INDEX 2 4 7 11 15 17 19 21 2
    &END QM_KIND
    &QM_KIND H
      MM_INDEX 5 6 8 9 10 12 13 16 18
    &END QM_KIND
  &END QMMM
  &SUBSYS
    &CELL
      ABC 41.751 41.751 41.751
```

QMMM

QM Cell

QM atoms

```
  &SUBSYS
    &CELL
      ABC 41.751 41.751 41.751
      ALPHA_BETA_GAMMA 90 90 90
      PERIODIC XYZ
    &END CELL
    &KIND H
      BASIS_SET DZVP-MOLOPT-GTH
      ELEMENT H
      POTENTIAL GTH-BLYP-q1
    &END KIND
    &KIND C
```

Total Cell

BASIS SET

```
      POTENTIAL GTH-BLYP-q5
    &END KIND
    &TOPOLOGY
      CONN_FILE_FORMAT AMBER
      CONN_FILE_NAME MQAE.prmtop
      COORD_FILE_FORMAT PDB
      COORD_FILE_NAME MQAE.pdb
    &END TOPOLOGY
  &END SUBSYS
&END FORCE_EVAL
```

FF Data

# Topology

**CONN_FILE_FORMAT**: *enum* = *GENERATE*

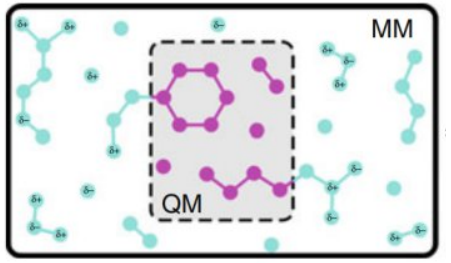
**Aliases:** CONNECTIVITY

**Usage:** *CONN_FILE_FORMAT (PSF|UPSF|MOL_SET|GENERATE|OFF|G87|G96|AMBER|USER)*

**Valid values:**

- `PSF` Use a PSF file to determine the connectivity. (support standard CHARMM/XPLOR and EXT CHARMM)
- `UPSF` Read a PSF file in an unformatted way (useful for not so standard PSF).
- `MOL_SET` Use multiple PSF (for now...) files to generate the whole system.
- `GENERATE` Use a simple distance criteria. (Look at keyword BONDPARM)
- `OFF` Do not generate molecules. (e.g. for QS or ill defined systems)
- `G87` Use GROMOS G87 topology file.
- `G96` Use GROMOS G96 topology file.
- `AMBER` Use AMBER topology file for reading connectivity (compatible starting from AMBER V.7)
- `USER` Allows the definition of molecules and residues based on the 5th and 6th column of the COORD section. This option can be handy for the definition of molecules with QS or to save memory in the case of very large systems (use PARA_RES off).

Ways to determine and generate a molecules. Default is to use GENERATE [Edit on GitHub]




**COORD_FILE_FORMAT**: *enum* = *OFF*

**Aliases:** COORDINATE

**Usage:** *COORD_FILE_FORMAT (OFF|PDB|XYZ|G96|CRD|CIF|XTL|CP2K)*

**Valid values:**

- `OFF` Coordinates read in the &COORD section of the input file
- `PDB` Coordinates provided through a PDB file format
- `XYZ` Coordinates provided through an XYZ file format
- `G96` Coordinates provided through a GROMOS96 file format
- `CRD` Coordinates provided through an AMBER file format
- `CIF` Coordinates provided through a CIF (Crystallographic Information File) file format
- `XTL` Coordinates provided through a XTL (MSI native) file format
- `CP2K` Read the coordinates in CP2K &COORD section format from an external file. NOTE: This file will be overwritten with the latest coordinates.

Set up the way in which coordinates will be read. [Edit on GitHub]

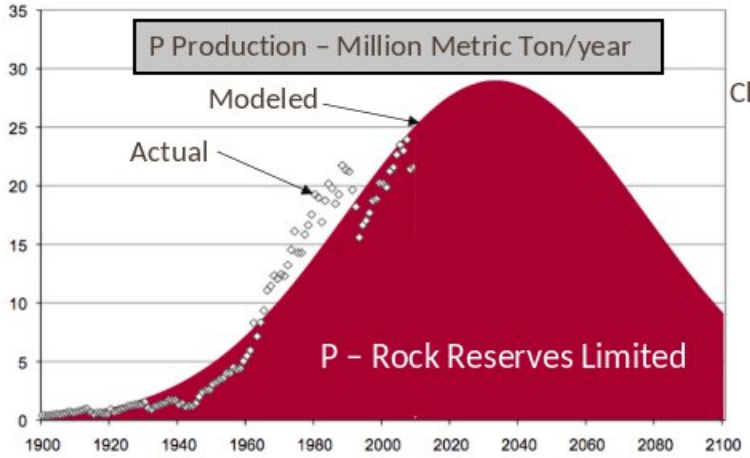
```
      POTENTIAL GTH-BLYP-q5
    &END KIND
    &TOPOLOGY
      CONN_FILE_FORMAT AMBER
      CONN_FILE_NAME MQAE.prmtop
      COORD_FILE_FORMAT PDB
      COORD_FILE_NAME MQAE.pdb
    &END TOPOLOGY
  &END SUBSYS
&END FORCE_EVAL
```

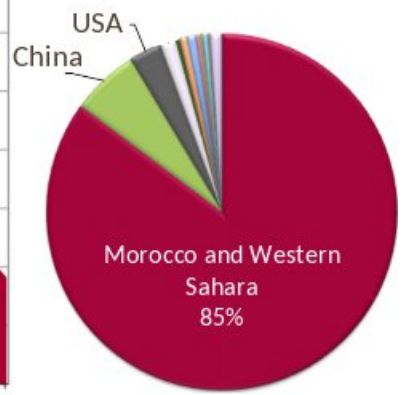
Inputfile

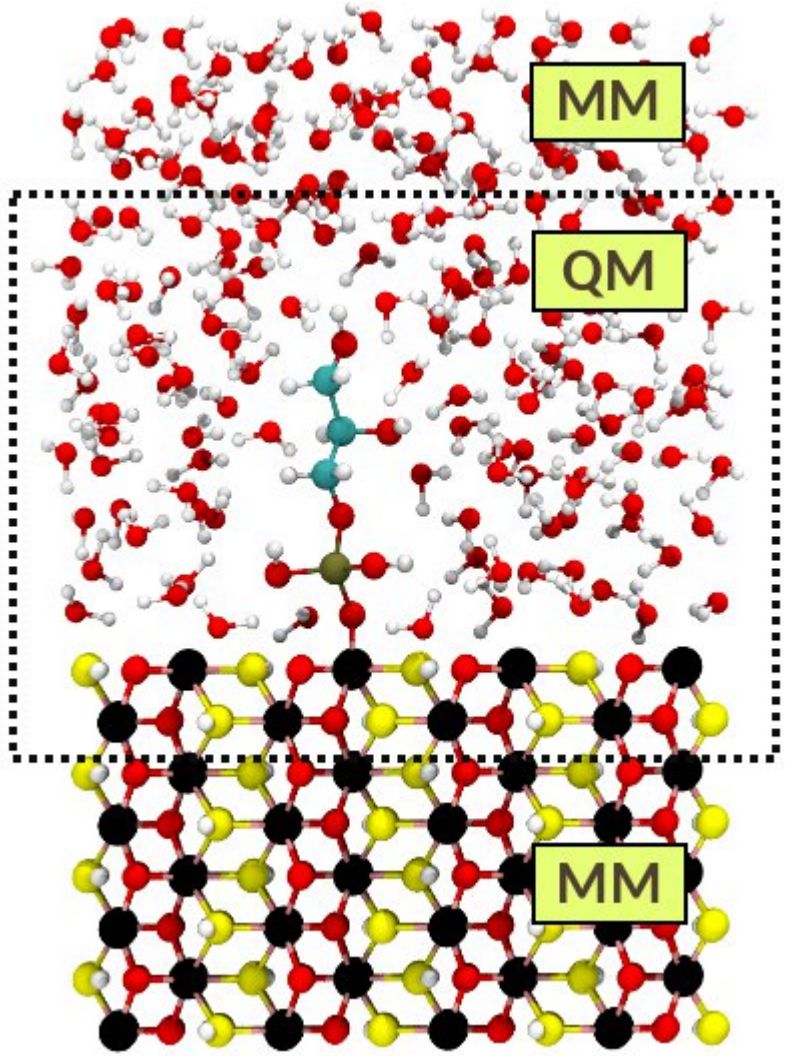
# Example 2

## P Peak Scenario

- Peak P scenario could happen soon
- Oil Crisis in 1970; N0 alternative for P



# Benchmarks

1picosecond simulation of
1000 water molecules (3000 atoms
in 1 and half day.

